AdvancedRNA-seq

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##I used the markdown from week 7, and edited it according to the extra help markdown for the ICA. The file mytable_feaures was taken from alignment files. The annotation file was created using the information on NCBI (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE229693)

All of the code here is taken from class material of Functional Genomic Technologies Week 6 or Example Analysis Extra ICA document unless stated otherwise. Code blocks that are taken from Example Analysis Extra ICA document are stated. If not stated, the code were taken from Week 6 RNA-Seq Tutorial Files.

CHATGTP was used to debug the code or further understand error messages. ## Analysis of RNA-seq data using R

- Create a directory to store this week's files and unpack the archive file for tutorial7 into this folder

cp /shared_files/FGT_T7_RNAseq_tutorial_files.tar . tar -xvf FGT_T7_RNAseq_tutorial_files.tar • Start R • Load the libraries needed for processing (see below) • Load the R files for the tutorial- generated from the download, QC and alignment code (see below. • Run the rest of the analysis

```
#Load the required libraries & load the files for the tutorial
library(Rsubread)
library(edgeR)
```

```
## Loading required package: limma
```

```
library(limma)
library(gplots)
```

```
##
## Attaching package: 'gplots'
```

```
## The following object is masked from 'package:stats':
##
##     lowess
```

```
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following object is masked from 'package:limma':
##
##     plotMA
```

```
## The following objects are masked from 'package:stats':
```

```
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, setdiff, table,
##     tapply, union, unique, unsplit, which.max, which.min

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:gplots':
##
##     space

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges

## Loading required package: GenomicRanges

## Loading required package: GenomeInfoDb

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars

## Loading required package: Biobase
```

```
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.

##
## Attaching package: 'Biobase'

## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians

## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(affy)
library(QuasR)
```

```
## Loading required package: parallel

## Loading required package: Rbowtie
```

```
library(pheatmap)
load("mytable_feaures")
#We now have files for the tutorial and also data loaded into the workspace-
#use ls() and dir() to view this data
ls()
```

```
## [1] "mytable_feaures"
```

```
dir()
```

```
##  [1] "_STARtmp"                          "Aligned.out.sam"
##  [3] "alignment"                         "FGTICA_targets.txt"
##  [5] "GSE229693_multiqc_report_2.html" "GSE229693_multiqc_report.html"
##  [7] "GSE229693.csv"                     "heatmap_candidates_ICA.png"
##  [9] "heatmap_candidates.png"            "ICA_functional.Rproj"
## [11] "ICa.pdf"                           "ICa.Rmd"
## [13] "Log.out"                           "Log.progress.out"
## [15] "mouse_H_v5p2.rdata"                "mytable_feaures"
## [17] "results_genotype.csv"              "volcano_test_2.png"
## [19] "volcano_test_3.png"                "volcano_test.png"
```

Once we have loaded the data we are ready to run some further analysis.

```
#Some further processing to access count data and column names.
table_rnaseq<-(mytable_feaures)$counts
colnames_rnaseq <-colnames(table_rnaseq)
#tidy up the sample names eg"X.home.guest101.new_rnaseq.aligned.SRR5054353.sorted.bam"
#to ShortName  using the " targets file called adf here
adf <-read.table("FGTICA_targets.txt",sep='\t', row.names=1,fill=T,header=T)
#This orders the elements in the table
idx<-match(colnames_rnaseq,rownames(adf))
colnames(table_rnaseq) <-adf$ShortName[idx]
adf <-adf[idx,]
rownames(adf) <-colnames(table_rnaseq)
```

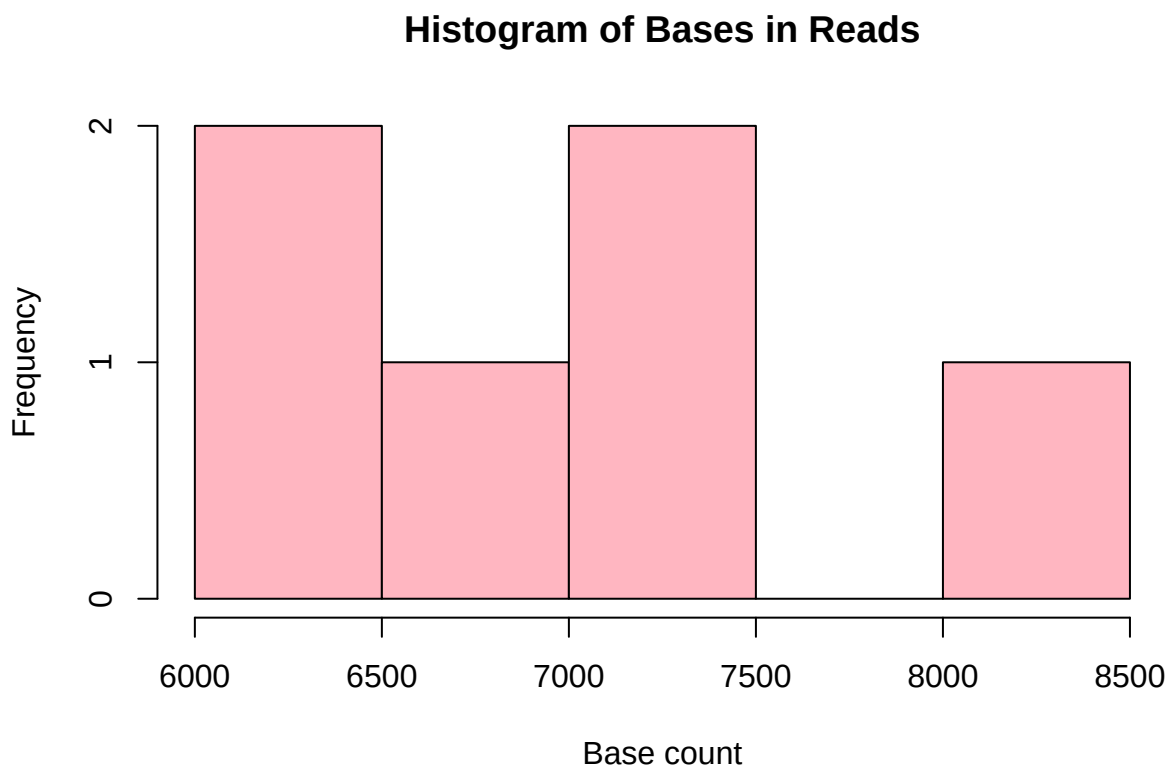
Exploring the Sample Annotation

We can use R to summarise the annotation table from this study.

```
#Load the annotation table- here it is a data frame
adf <-read.table("FGTICA_targets.txt",sep='\t', row.names=1,fill=T,header=T)
#The annotation table contains data in a number of named columns
colnames(adf)
```

```
##  [1] "BioSample"     "Experiment"    "SampleName"    "Source_name"
##  [5] "GEO_Accession" "AVGSpotLEn"    "Bases.M."      "Bytes.Mb."
##  [9] "ShortName"     "Line"          "Genotype"
```

```
#We can generate a histogram to explore the library sizes in this data.
#All the codes regarding the histograms were edited with guidance from
#https://www.rdocumentation.org/packages/graphics/versions/3.6.2/topics/hist
hist(as.numeric(adf$Bases.M.),
     main="Histogram of Bases in Reads",
     xlab= "Base count",
     ylab="Frequency",
     col="lightpink"
     )
```




```
#It is also interesting to summarise the table in terms of factors called "Line" and "Sex".
table(adf[,c("Source_name","Genotype")])
```

```
##                       Genotype
## Source_name            floxed WBSCR16_KO
##   brown adipose tissue      3          3
```

Counting Genomic Features

We now have reads mapped to the genome, gene annotation files but the reads are not mapped and counted within each annotation region (eg within each gene). The reads aligned to the genome are in a file format called BAM see https://www.ncbi.nlm.nih.gov/sra/docs/submitformats/. If we were to view this file we would see the regions to which the reads map, but in the next stage of the pipeline we use a tool called featureCounts to count the reads that map within each region we are interested in. In the interest of time, the command is provided but commented out. The results of running featureCounts were loaded into R when we loaded the serialised ".Rdata" object for this experiment.

```
#use featureCounts to load the BAM files into the
#mytable_feaures <-featureCounts(files=alignment_files,annot.ext=".
#/omes/annot/Mus_musculus.GRCm38.99.gtf",isGTFAnnotationFile = TRUE, countMultiMappingReads=F, minMQS=3
#nthreads=10,minOverlap=10)
#feature counts takes the genome annotation file #(Mus_musculus.GRCm38.99.gtf) and the
#BAM files themselves.  Note #that alignment_files is just a list of file locations.
#Check the contents of the alignment_files
#alignment_files
#Note also that we have countMultiMappingReads =F this means that #reads that map to more than one geno
#minMQS - minimum mapping quality score,30 is a very stringent value #(we return to this later)
#nthreads =10 run ten threads at once (another reason to pre-compute #this in a large class!
#minOverlap =10 - the minimum overlap between the feature and the read- here 10bp
#Some further processing to access count data and column names.
table_rnaseq<-(mytable_feaures)$counts
colnames_rnaseq <-colnames(table_rnaseq)
#tidy up the sample names #eg fix #"X.home.guest101.new_rnaseq.aligned.SRR5054353.sorted.bam"!
#to ShortName using the " targets file" object called adf here
#This orders the elements in the table
idx<-match(colnames_rnaseq,rownames(adf))
colnames(table_rnaseq) <-adf$ShortName[idx]
adf <-adf[idx,]
rownames(adf) <-colnames(table_rnaseq)
#Finally check the output of the reorganisation
head(table_rnaseq)
```

```
##                     FL1 FL2 FL3 KO1 KO2 KO3
## ENSMUSG00000104478   0   0   0   0   0   0
## ENSMUSG00000104385   0   0   0   0   0   0
## ENSMUSG00000101231   0   0   0   0   0   0
## ENSMUSG00000102135   4   1   0   2   2   0
## ENSMUSG00000103282   0   0   0   0   0   0
## ENSMUSG00000101097   0   0   0   0   0   0
```

Removal of Unwanted Samples

Notice from the annotation we have two XO samples, we can convert the design to a two factor factorial design by removing these two samples. We then have three samples in each combination of the two factors. - I did not run this code block as my study only had one condition.

```
#Adjust sample data
#table(adf[,c("Line","Sex")])
#keep-remove XO samples
#keep_cols=adf$Sex[idx]!="O"
#We can use this to remove the unwanted columns from the adf and #also the data eg
#table(adf[,c("Line","Sex")])
#With just the required columns
```

```
#table(adf[keep_cols,c("Line","Sex")])
#adf <-adf[keep_cols,]
```

Loading the Data into DESeq

DESeq2 will used for the statistical analysis and the normalisation of the data.

```
#Create object for DESeq
#NOTE the design for differential analysis is fed into this object
dds_rnaseq <- DESeqDataSetFromMatrix(countData = table_rnaseq,colData = adf,design = ~ Genotype)
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```
#check the object dimensions
dim(dds_rnaseq)
```

```
## [1] 78298     6
```

```
head(rownames(dds_rnaseq))
```

```
## [1] "ENSMUSG00000104478" "ENSMUSG00000104385" "ENSMUSG00000101231"
## [4] "ENSMUSG00000102135" "ENSMUSG00000103282" "ENSMUSG00000101097"
```

```
colnames(dds_rnaseq)
```

```
## [1] "FL1" "FL2" "FL3" "KO1" "KO2" "KO3"
```

```
#check the object type
class(dds_rnaseq)
```

```
## [1] "DESeqDataSet"
## attr(,"package")
## [1] "DESeq2"
```

```
typeof(dds_rnaseq)
```

```
## [1] "S4"
```

```
#check what this object contains
slotNames(dds_rnaseq)
```

```
## [1] "design"             "dispersionFunction" "rowRanges"
## [4] "colData"            "assays"             "NAMES"
## [7] "elementMetadata"    "metadata"
```

```
colnames(colData(dds_rnaseq))
```

```
##  [1] "BioSample"     "Experiment"    "SampleName"    "Source_name"
##  [5] "GEO_Accession" "AVGSpotLEn"    "Bases.M."      "Bytes.Mb."
##  [9] "ShortName"     "Line"          "Genotype"
```

```
colnames(assay(dds_rnaseq))
```

```
## [1] "FL1" "FL2" "FL3" "KO1" "KO2" "KO3"
```

Quality Filtering Genes Apply filters to remove genes that do not pass quality filters.

```
#first filter based upon counts
keep <- rowSums(counts(dds_rnaseq)) > 1
```

```
dds_rnaseq <- dds_rnaseq[keep,]
nrow(dds_rnaseq)
```

```
## [1] 35996
```

```
#second filter we might apply-require >10 in >= 3 samples
#rowSums here counts the number of TRUE elements per row
# counts(dds_rnaseq) >= 10 returns TRUE if each row has >= 10 TRUE
#counts
keep <- rowSums(counts(dds_rnaseq) >= 10) >= 3
dds_rnaseq <- dds_rnaseq[keep,]
#We now have a DESeq object filtered for samples and genes that pass #QC
nrow(dds_rnaseq)
```

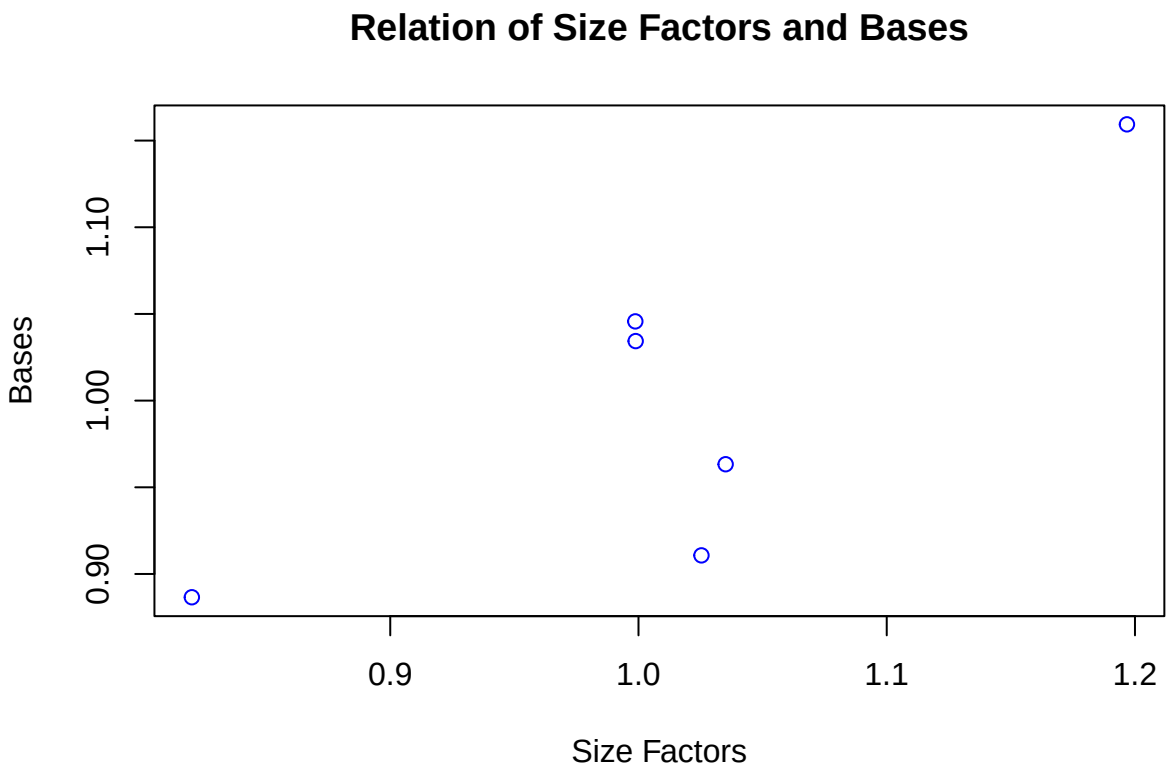
```
## [1] 16659
```

Normalisation of Data

```
#size factor estimation for normalisation- these are used to
#scale by library size when data is exported.
dds_rnaseq <- estimateSizeFactors(dds_rnaseq)
sizeFactors(dds_rnaseq)
```

```
##       FL1       FL2       FL3       KO1       KO2       KO3
## 0.9987165 0.8200767 0.9988631 1.0351099 1.1967659 1.0253562
```

```
#The size factors should relate to the library size.  We can check this from the annotation
adf_bases <- adf$Bases.M.
names(adf_bases) <- rownames(adf)
adf_bases <- adf_bases/mean(adf_bases)
#Check how these two tables relate
#Code was edited with guidance of
#https://www.rdocumentation.org/packages/graphics/versions/3.6.2/topics/plot
plot(sizeFactors(dds_rnaseq), adf_bases,
     main="Relation of Size Factors and Bases",
     xlab="Size Factors",
     ylab="Bases",
     col="blue")
```



DESeq2 also provides methods that outputs data with various normalisations.

Here we generate multiple different normalisations and compare between then- but for the rest of the tutorial we use rlog normalisation! Note that the normalised data is not used for differential expression analysis- just for data visualisation.

```
#Normalised log counts
counts_rnaseq <-log2(counts(dds_rnaseq, normalized=TRUE))
#fpm is similar to cpm and discussed here
#https://rdrr.io/bioc/DESeq2/man/fpm.html.  Here the data is log transformed
fpm_rnaseq <-log(fpm(dds_rnaseq))#this is also normalised by the sizeFactors
#rlog ad vsd are discussed here
#https://genomebiology.biomedcentral.com/articles/10.1186/s13059-014-0550-8
vsd_rnaseq <- vst(dds_rnaseq, blind = T)
head(vsd_rnaseq, 3)
```

```
## class: DESeqTransform
## dim: 3 6
## metadata(1): version
## assays(1): ''
## rownames(3): ENSMUSG00000051285 ENSMUSG00000103509 ENSMUSG00000057363
## rowData names(4): baseMean baseVar allZero dispFit
## colnames(6): FL1 FL2 ... KO2 KO3
## colData names(12): BioSample Experiment ... Genotype sizeFactor
```

```
rld_rnaseq <- rlog(dds_rnaseq, blind = T)
head(rld_rnaseq, 3)
```

```
## class: DESeqTransform
## dim: 3 6
## metadata(1): version
## assays(1): ''
## rownames(3): ENSMUSG00000051285 ENSMUSG00000103509 ENSMUSG00000057363
## rowData names(7): baseMean baseVar ... dispFit rlogIntercept
## colnames(6): FL1 FL2 ... KO2 KO3
## colData names(12): BioSample Experiment ... Genotype sizeFactor
```

We will use rlog for the rest of this tutorial, but here we compare the different normalised data

##Plots with Normalised Data.

```
#Code regarding boxplots were edited with guidance of
#https://www.rdocumentation.org/packages/graphics/versions/3.6.2/topics/boxplot
boxplot(counts_rnaseq,
        main="Boxplot of Normalised Counts",
        ylab="Values",
        xlab="Samples",
        col="lightpink")
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 1 is not drawn
```

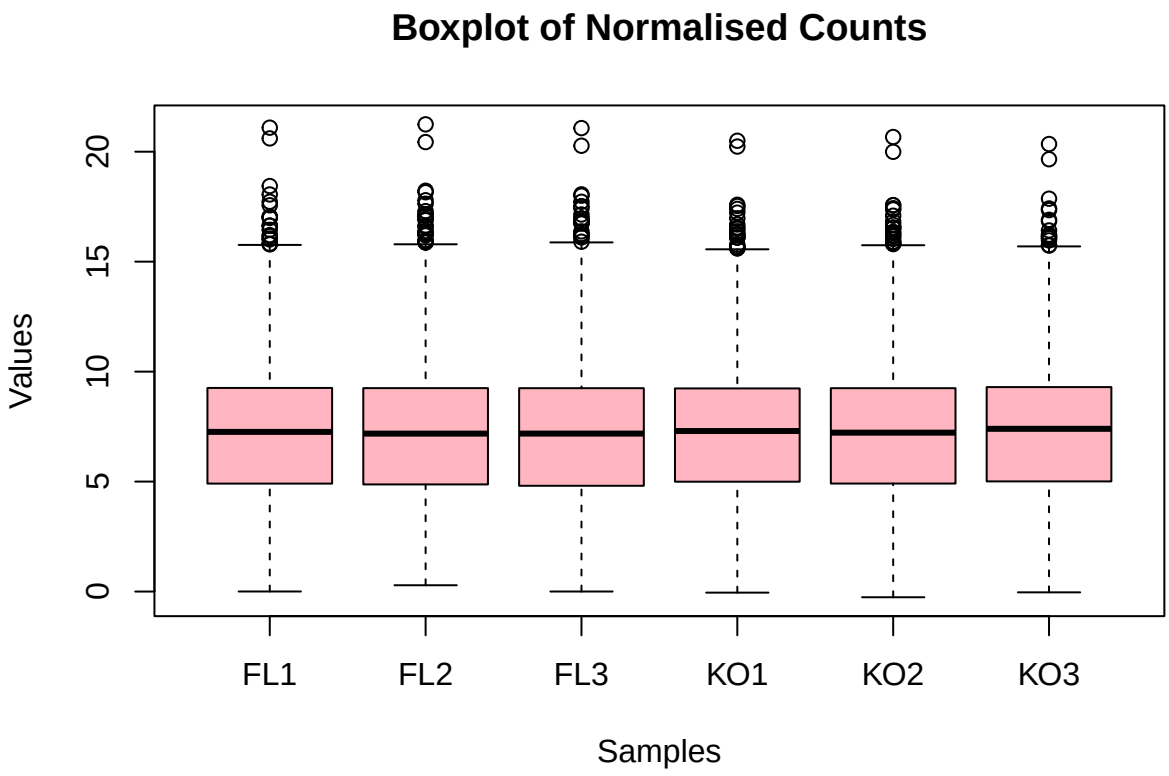
```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 2 is not drawn
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 3 is not drawn
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 4 is not drawn
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 5 is not drawn
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 6 is not drawn
```




```
boxplot(fpm_rnaseq,
        main="Boxplot of Normalised Counts (FPM)",
        ylab="Values",
        xlab="Samples",
        col="lightpink")
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 1 is not drawn

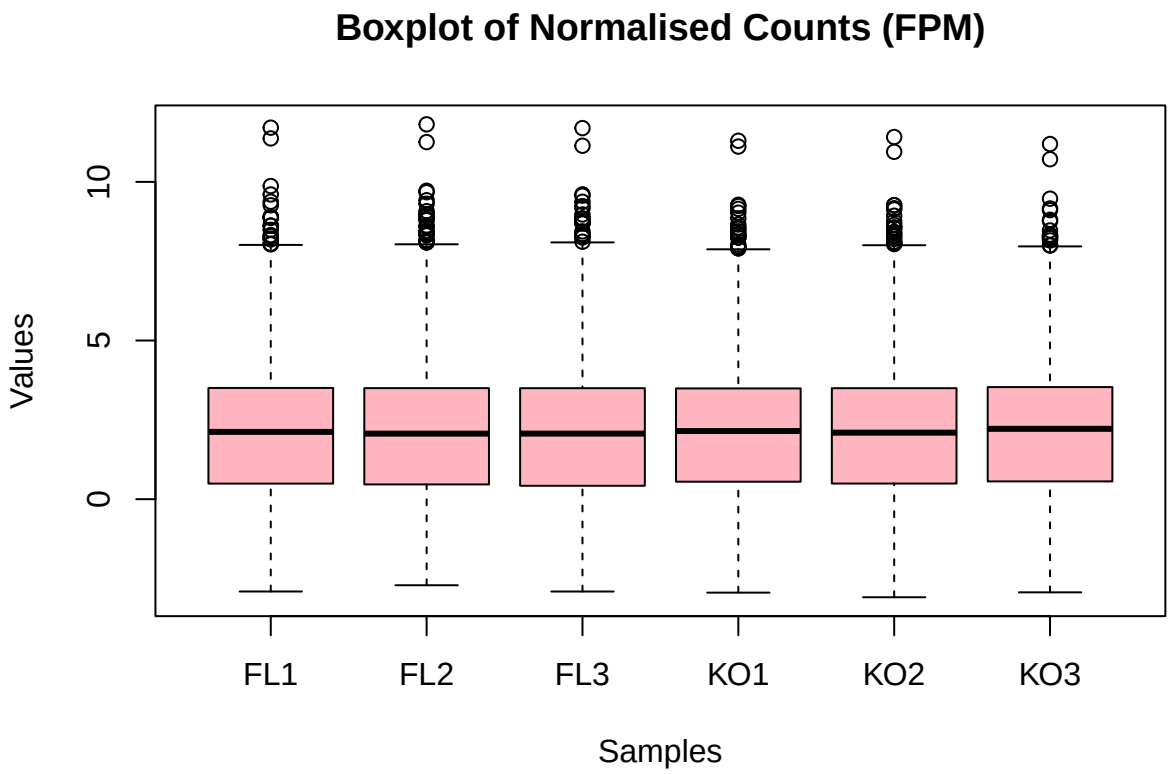
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 2 is not drawn

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 3 is not drawn

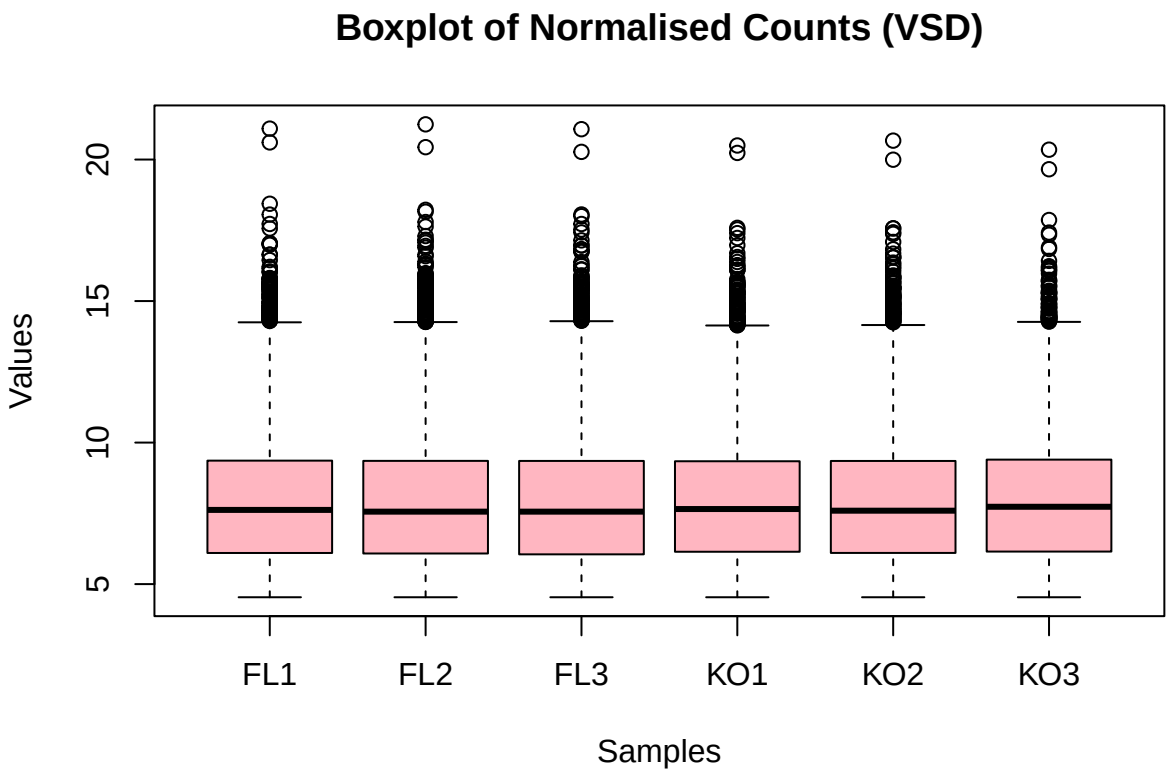
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 4 is not drawn

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 5 is not drawn

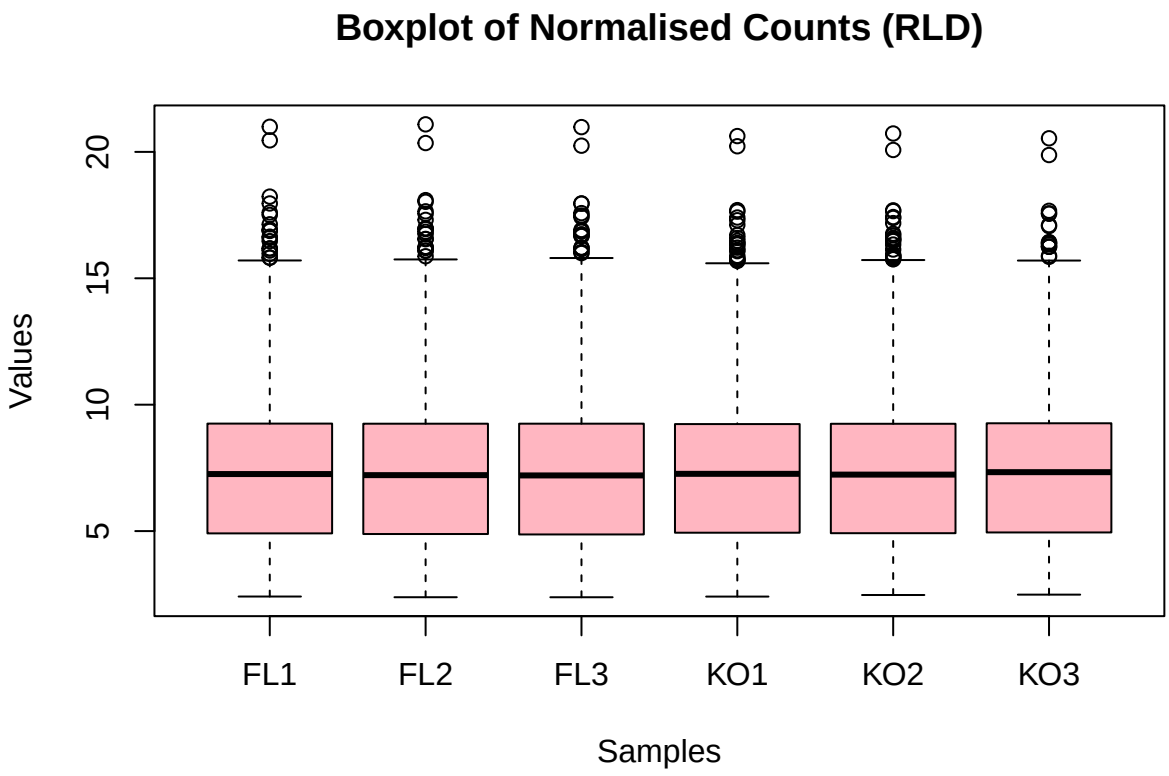
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 6 is not drawn
```




```
boxplot(assay(vsd_rnaseq),
        main="Boxplot of Normalised Counts (VSD)",
        ylab="Values",
        xlab="Samples",
        col="lightpink")#etc
```




```
boxplot(assay(rld_rnaseq),
        main="Boxplot of Normalised Counts (RLD)",
        ylab="Values",
        xlab="Samples",
        col="lightpink")#etc
```




```
#but suppose we look at non-normalised

na.rm=T
counts_un <-counts(dds_rnaseq, normalized=FALSE)
boxplot(log2(counts_un),
        main="Boxplot of Non-Normalised Counts without Log2",
        ylab="Values",
        xlab="Samples",
        col="lightpink")
```

```
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 1 is not drawn

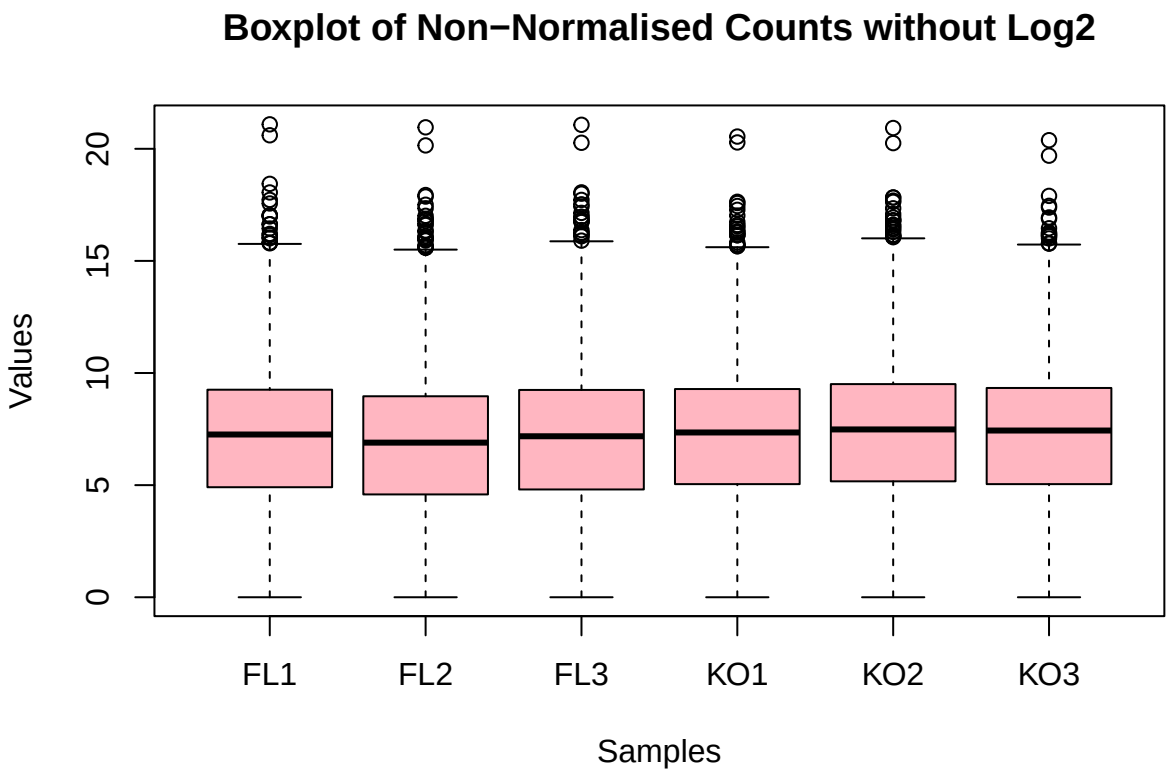
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 2 is not drawn

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 3 is not drawn

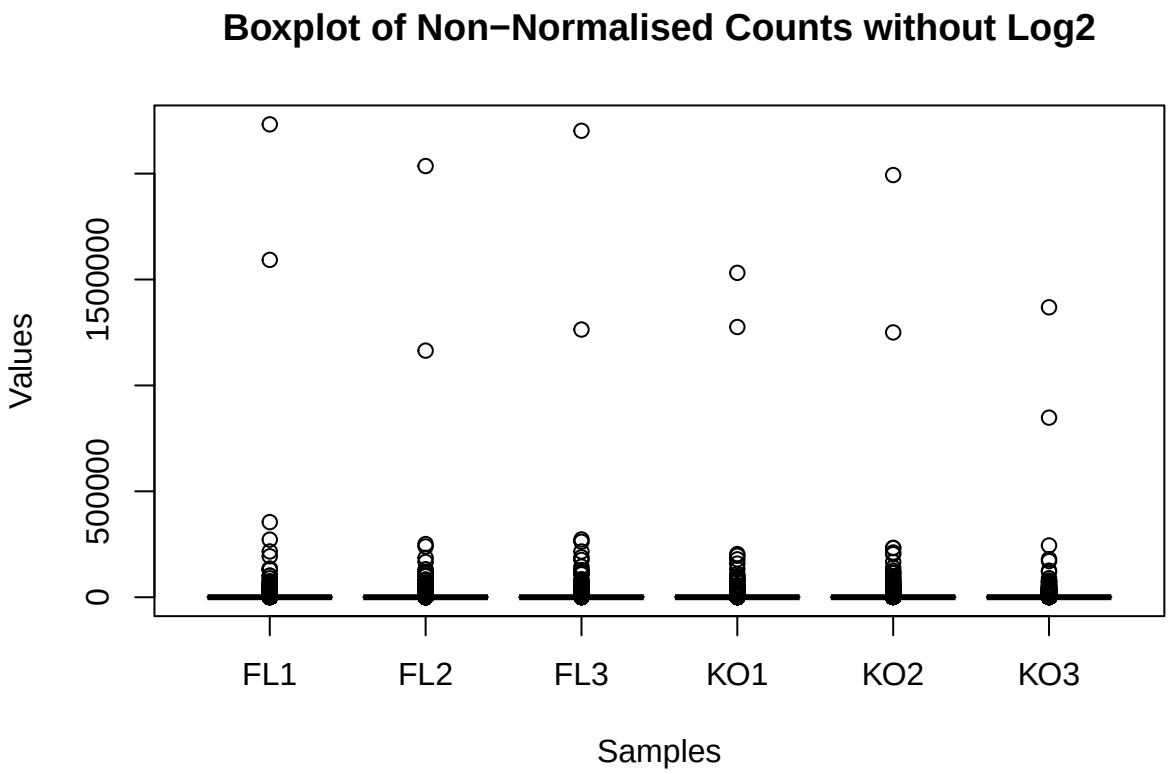
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 4 is not drawn

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 5 is not drawn

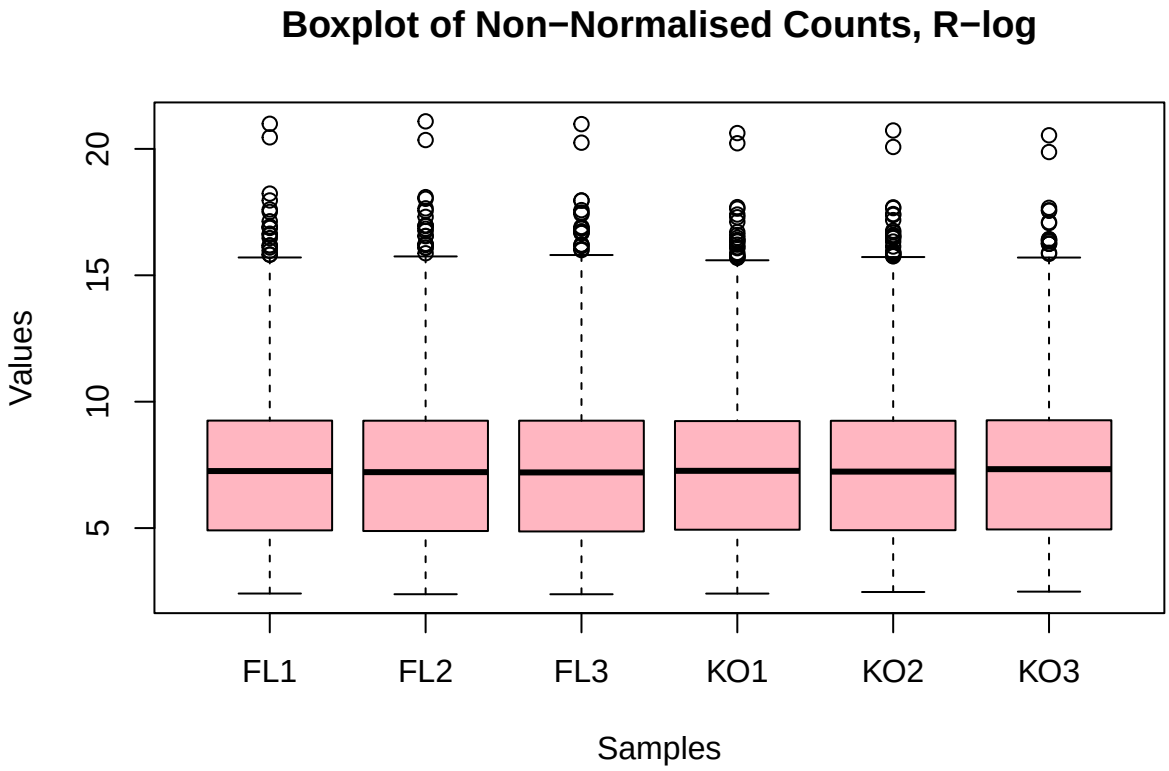
## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (-Inf) in boxplot 6 is not drawn
```




```
#I wanted to see how it would change without the log2 base, so I ran this too.
boxplot(counts_un,
        main="Boxplot of Non-Normalised Counts without Log2",
        ylab="Values",
        xlab="Samples",
        col="lightpink")
```




```
counts_un_rlog<-rlog(counts_un)
boxplot(counts_un_rlog,
        main="Boxplot of Non-Normalised Counts, R-log",
        ylab="Values",
        xlab="Samples",
        col="lightpink")
```

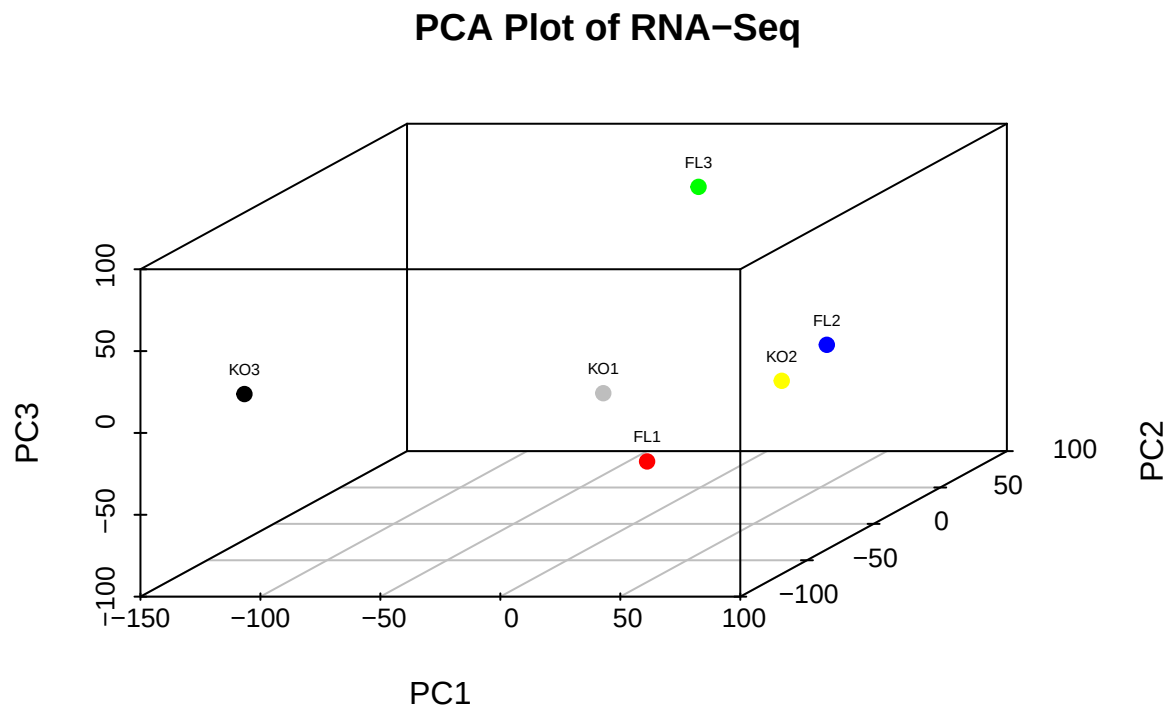



```
#We can make an mva.pairs plot for any of these data
#For example we can examine the unnormalized data.
#make mva.pairs-this is slow "mva.pairs.png" is provided
#png(filename="mva.pairs.png",width=2000, height=2000)
#mva.pairs(counts_un_rlog)
#dev.off()
```

Exploratory Data Plots PCA plots similar to those we use for microarrays.

```
#we can run a PCA plot here...
library(scatterplot3d)
library(ggplot2)
# Perform PCA
#Code was edited with guidance of
#https://cran.r-project.org/web/packages/ggfortify/vignettes/plot_pca.html
colours <- c("FL1"="red",
             "FL2"="blue",
             "FL3"="green",
             "KO1"="grey",
             "KO2"="yellow",
             "KO3"="black")
pca <- prcomp(t(na.omit(assay(rld_rnaseq))), scale=T)
# Plot the PCA results
# code edited with guidance from
#https://www.rdocumentation.org/packages/FactoMineR/versions/2.9/topics/PCA
s3d<-scatterplot3d(pca$x[,1:3], pch=19, color=colours, main="PCA Plot of RNA-Seq")
s3d.coords <- s3d$xyz.convert(pca$x[,1:3])
```

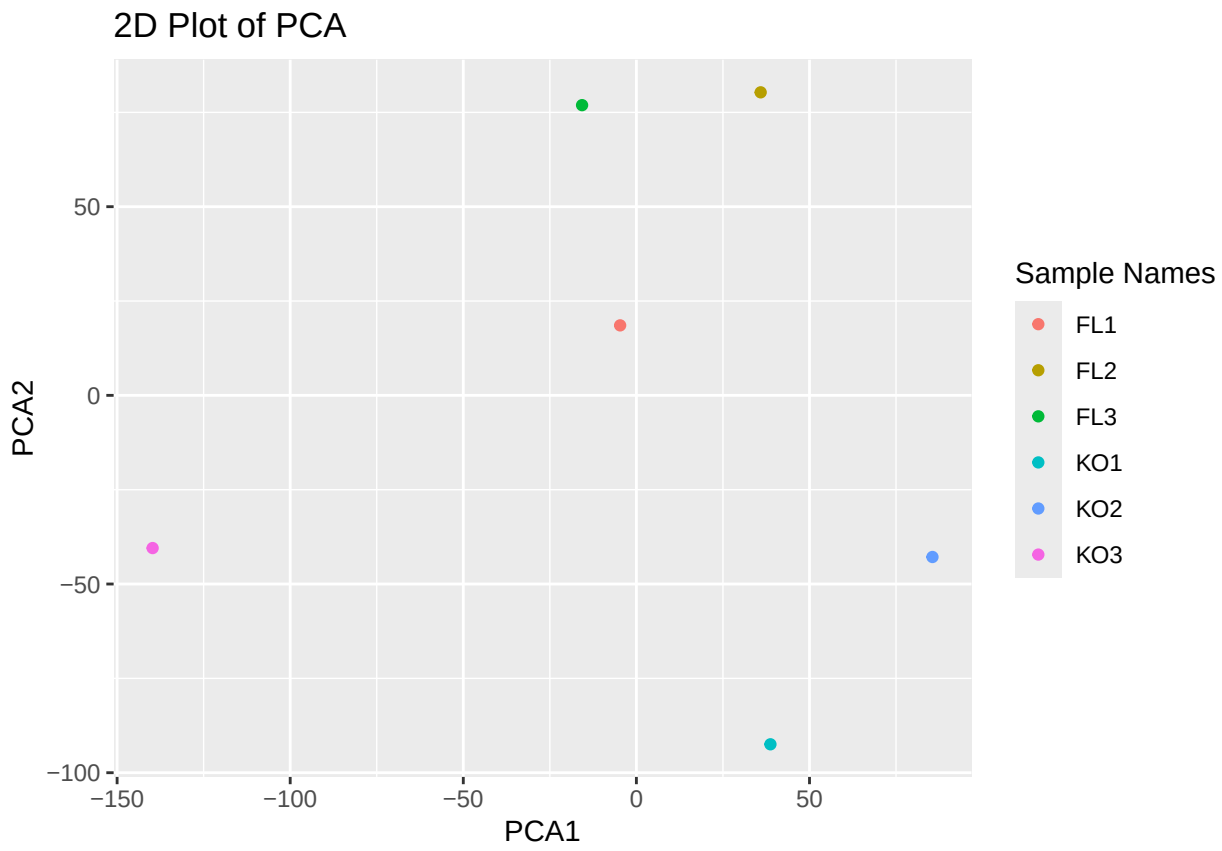
```
text(s3d.coords$x, s3d.coords$y,
     labels = colnames(rld_rnaseq),pos = 3,offset = 0.5,cex=0.5)
```



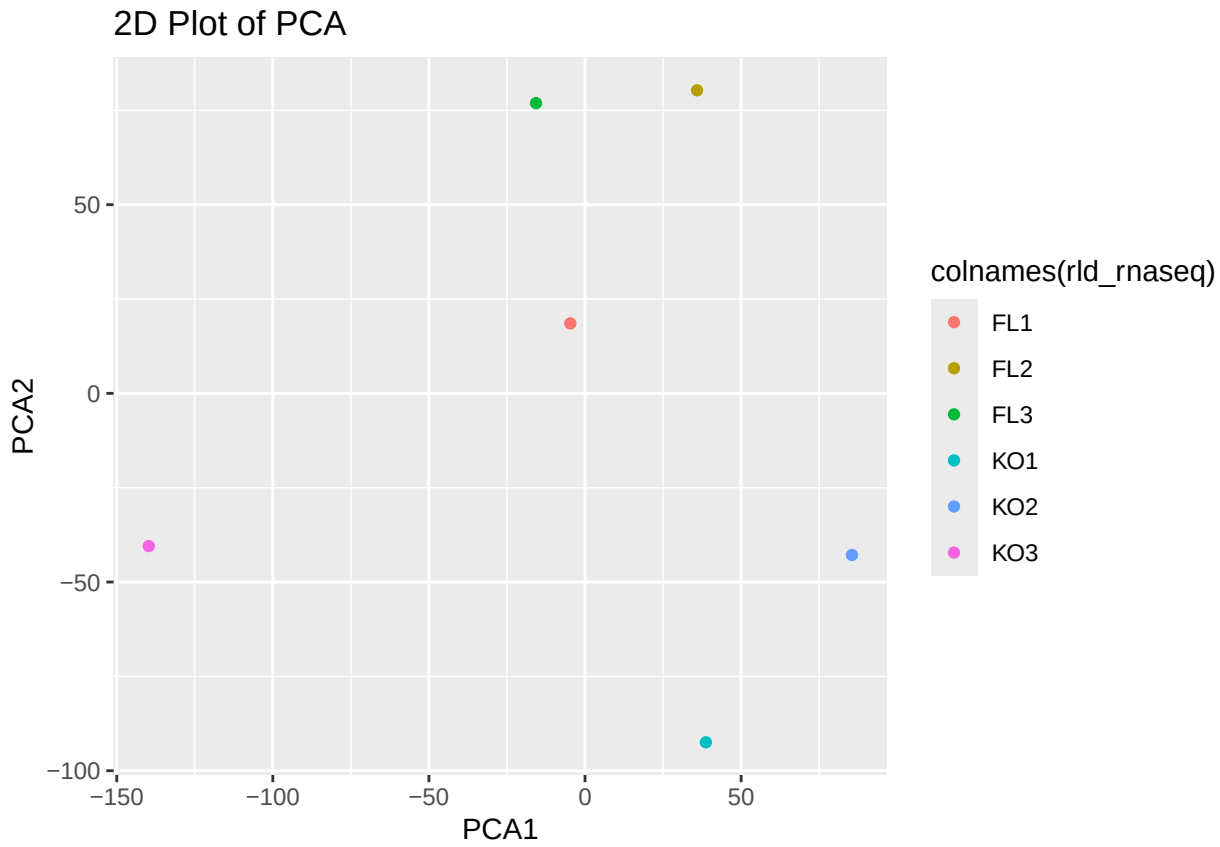

```
#or a quick plot in 2d, code was edited with guidance from
#https://support.bioconductor.org/p/9161338/ and
#https://stackoverflow.com/questions/76557653/cant-change-legend-title-of-ggplot
plotting <- qplot(pca$x[,1],pca$x[,2],main= "2D Plot of PCA",
                  xlab="PCA1", ylab="PCA2", col=colnames(rld_rnaseq))
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
plotting + labs(col = "Sample Names")
```


2D Plot of PCA
PCA2
PCA1
50
0
−50
−100
−150
−100
−50
0
50
Sample Names
FL1
FL2
FL3
KO1
KO2
KO3

```
plotting
```

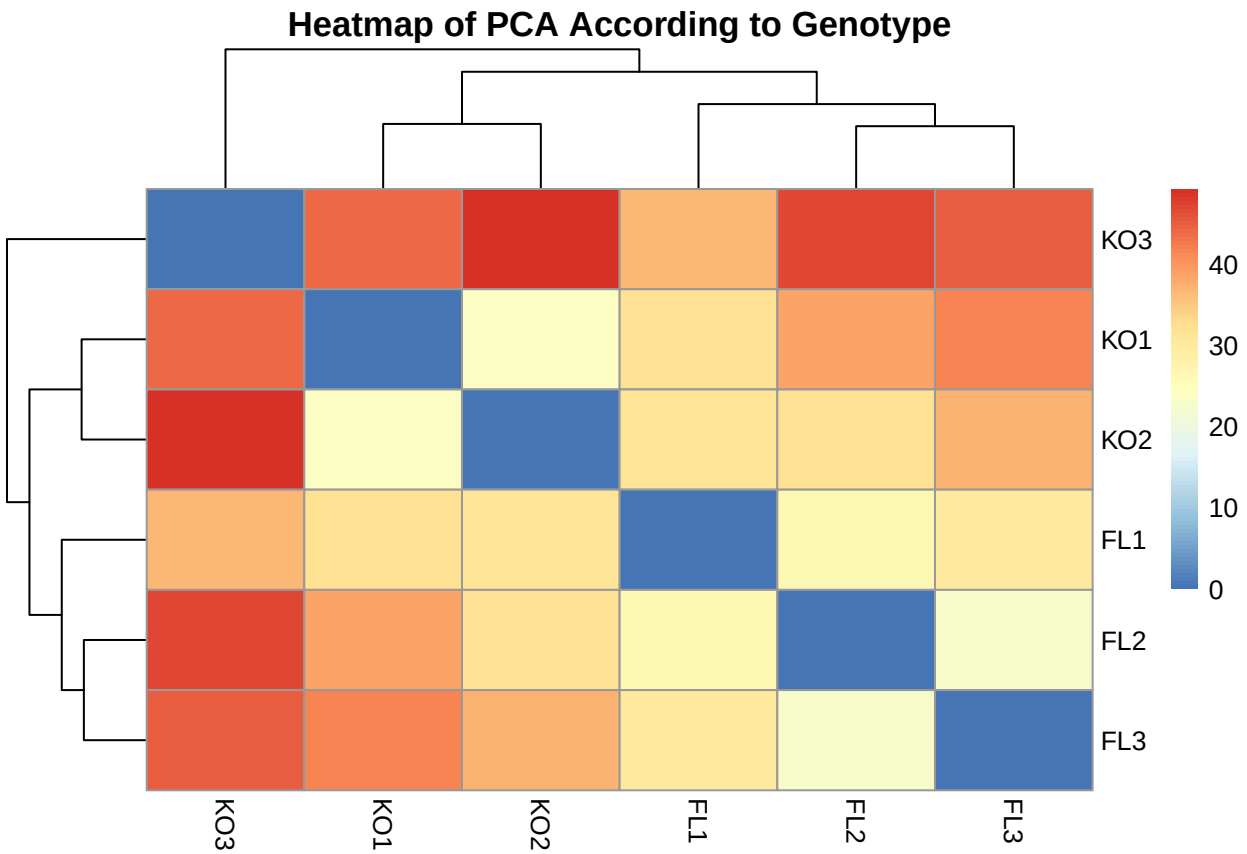



```
#We can also output a distance matrix...
sampleDists <- dist(t(assay(rld_rnaseq)))
sampleDists
```

```
##          FL1      FL2      FL3      KO1      KO2
## FL2 26.14885
## FL3 30.42686 23.12367
## KO1 32.15692 38.79669 41.73290
## KO2 31.37093 32.44964 37.34410 23.65720
## KO3 36.61084 47.00712 44.87447 43.98928 49.27060
```

Generate further heatmaps with the data

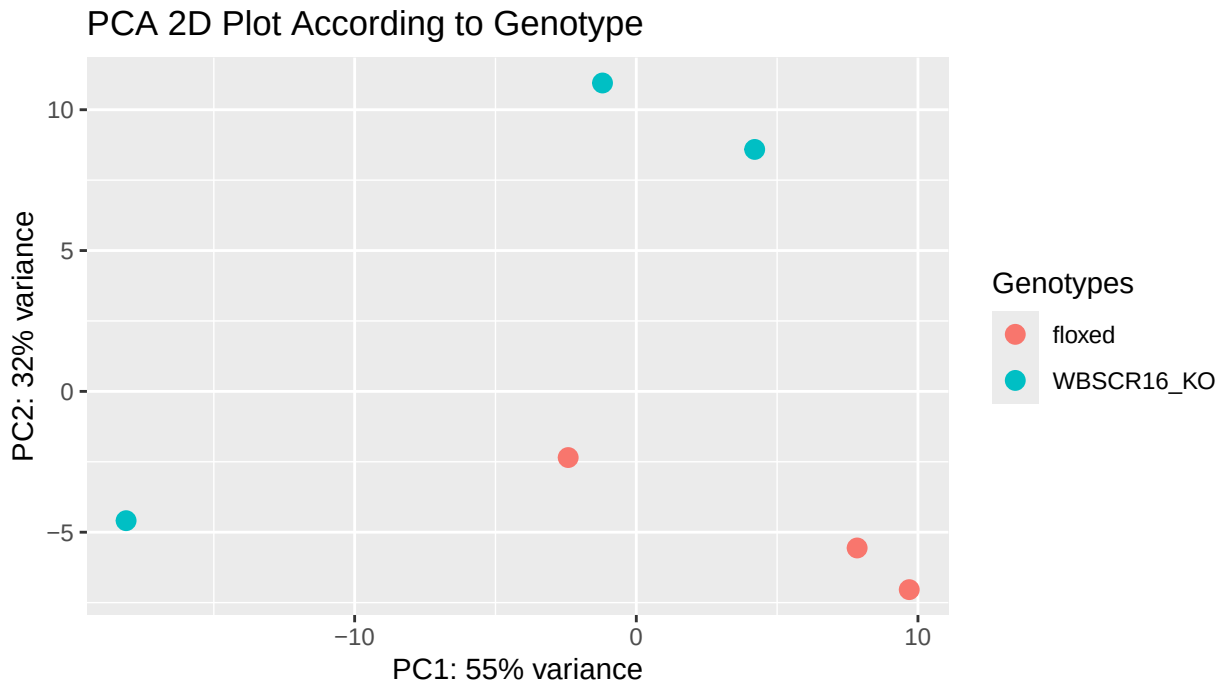
```
library("pheatmap")
library("RColorBrewer")
sampleDistMatrix <- as.matrix(sampleDists)
pheatmap(sampleDistMatrix, main="Heatmap of PCA According to Genotype")
```



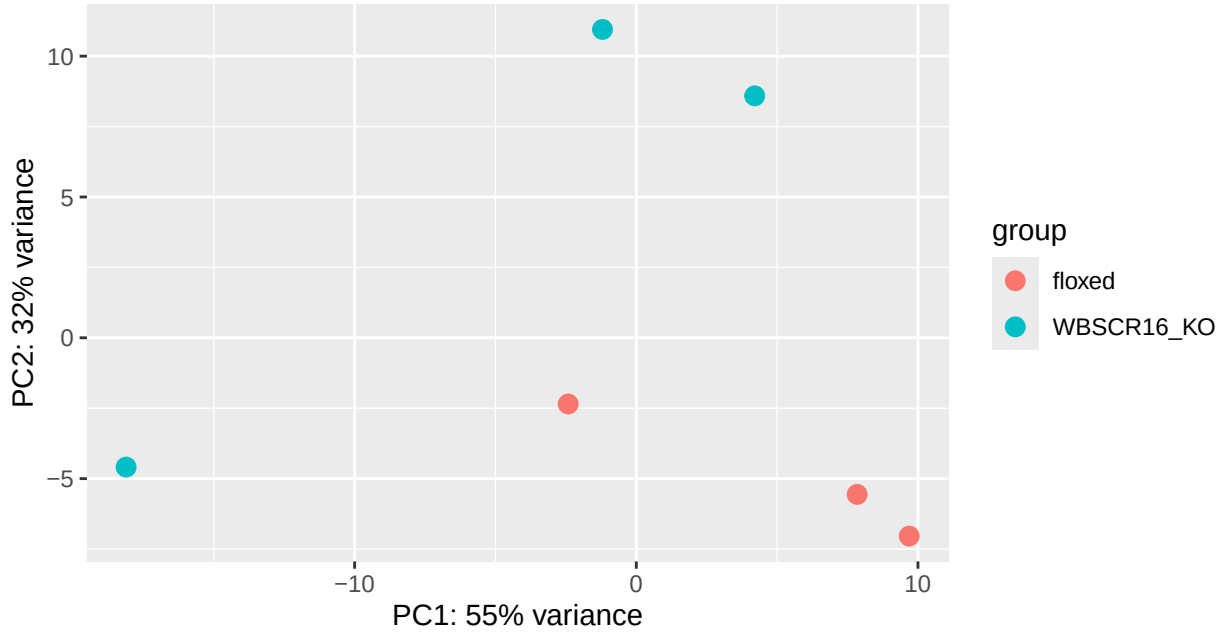

```
#easier PCA plot
#Code was edited with guidance from https://support.bioconductor.org/p/9161338/
#and https://stackoverflow.com/questions/76557653/cant-change-legend-title-of-ggplot
pcaplot <- plotPCA(rld_rnaseq, intgroup = c("Genotype"))
```

```
## using ntop=500 top features by variance
```

```
pcaplot +ggtitle("PCA 2D Plot According to Genotype")+ labs(col = "Genotypes")
```


PCA 2D Plot According to Genotype
PC2: 32% variance
PC1: 55% variance
10
5
0
−5
−10
0
10
Genotypes
floxed
WBSCR16_KO

```
pcaplot
```



Differential Gene Expression

Here we perform a two factor analysis using DESeq2. The design was fed into the dds_rnaseq object and was specified out as ~Line + Sex which specifies two factors Sex and Line. Sex and Line are colnames in the annotation able and samples are therefore annotated with these labels. Statistics in DESeq2 works in a similar way to Limma but is designed for count data.

```
dds_rnaseq <- DESeq(dds_rnaseq)
```

```
## using pre-existing size factors
```

```
## estimating dispersions
```

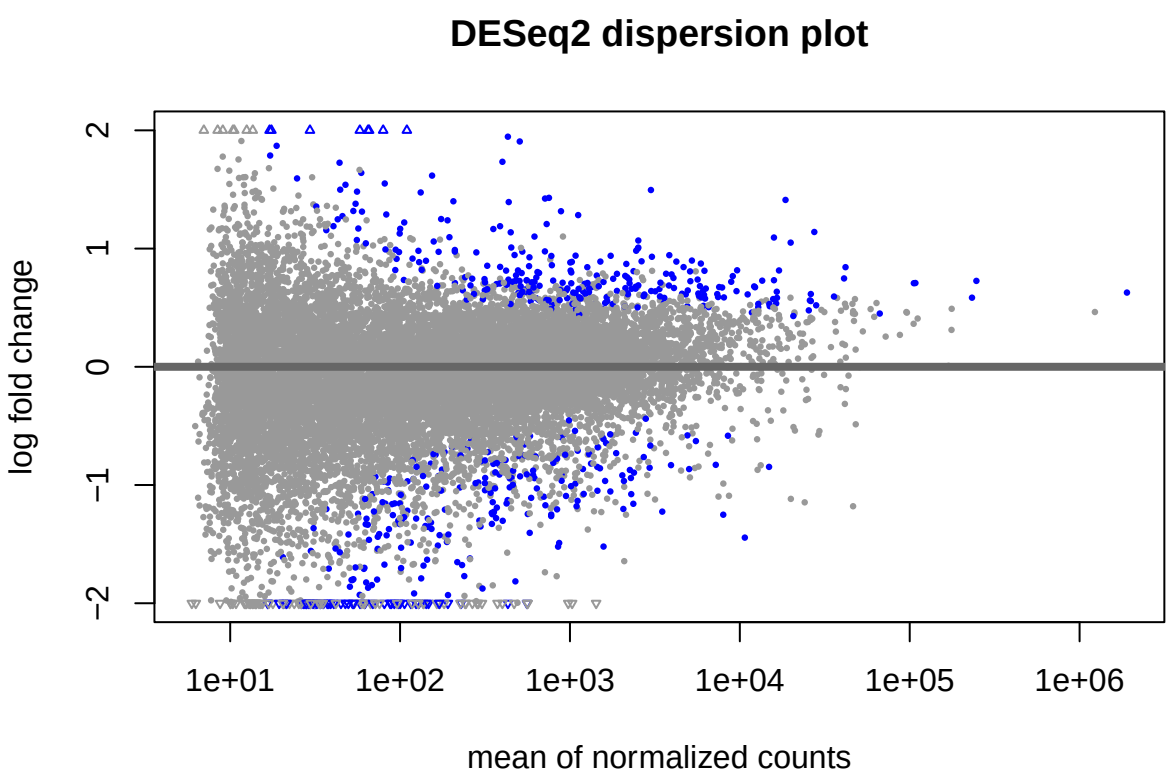
```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
#This is how you specify a contrast...
result_genotype <-results(dds_rnaseq,contrast=c("Genotype","floxed","WBSCR16_KO"))
#make a nice plot of differential expression
#Code was edited with guidance from
#https://www.rdocumentation.org/packages/DESeq2/versions/1.12.3/topics/plotMA
plotMA(result_genotype, main="DESeq2 dispersion plot", ylim=c(-2,2))
```




```
table(result_genotype$padj < 0.01)
```

```
##
## FALSE  TRUE
## 14278   103
```

```
table(result_genotype$padj < 0.05)
```

```
##
## FALSE  TRUE
## 14094   287
```

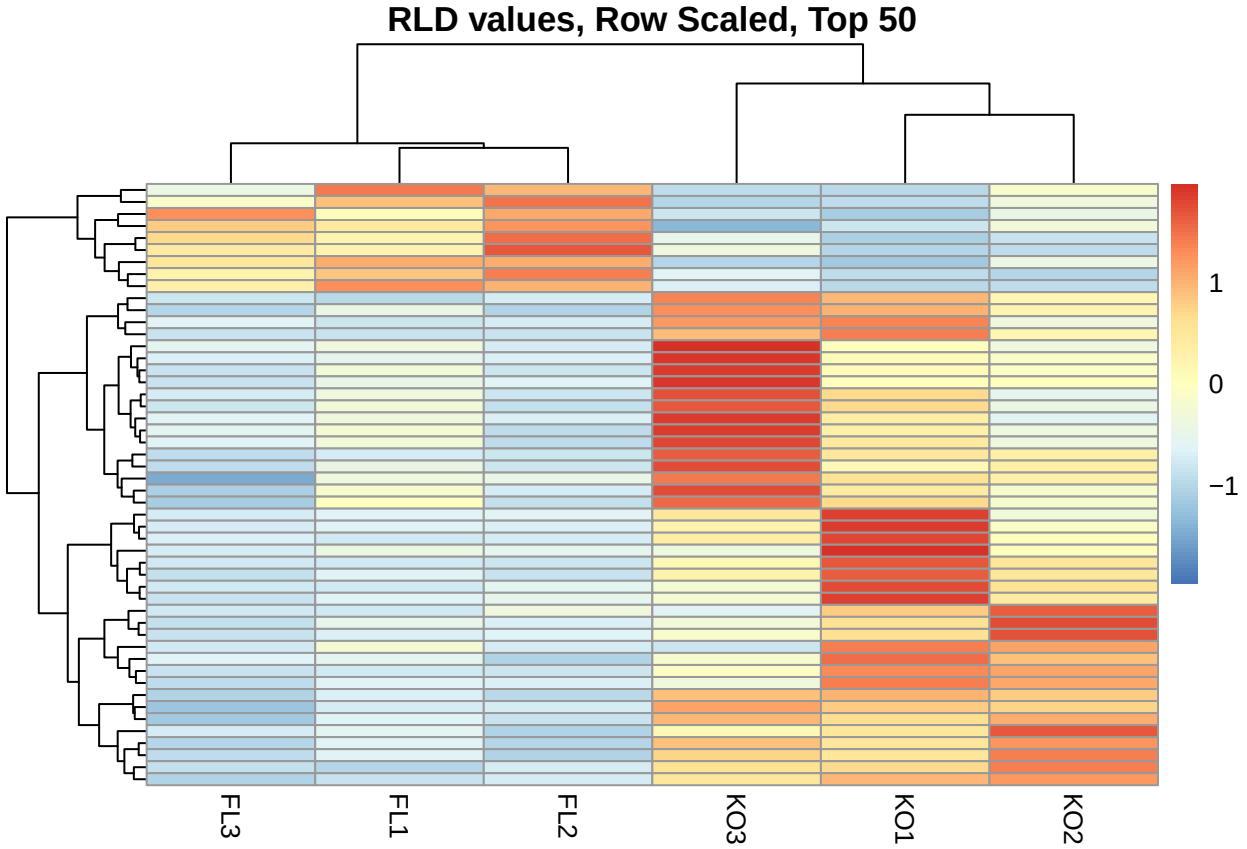
```
result_genotype_selected <- subset(result_genotype, padj < 0.05)
result_genotype_selected <-
  result_genotype_selected[order(abs(result_genotype_selected$log2FoldChange),
                                 decreasing = TRUE), ]
head(result_genotype_selected)
```

```
## log2 fold change (MLE): Genotype floxed vs WBSCR16_KO
## Wald test p-value: Genotype floxed vs WBSCR16_KO
## DataFrame with 6 rows and 6 columns
##                     baseMean log2FoldChange     lfcSE      stat      pvalue
##                    <numeric>      <numeric> <numeric> <numeric>   <numeric>
## ENSMUSG00000096410   49.6029       -7.15746   1.76103  -4.06435 4.81661e-05
## ENSMUSG00000094546   19.4191       -6.65256   1.35377  -4.91409 8.91974e-07
## ENSMUSG00000094689   21.4792       -5.90881   1.21472  -4.86433 1.14846e-06
## ENSMUSG00000093861   31.8639       -5.81094   1.15121  -5.04770 4.47161e-07
## ENSMUSG00000094420   38.0555       -5.77701   1.69441  -3.40946 6.50918e-04
```

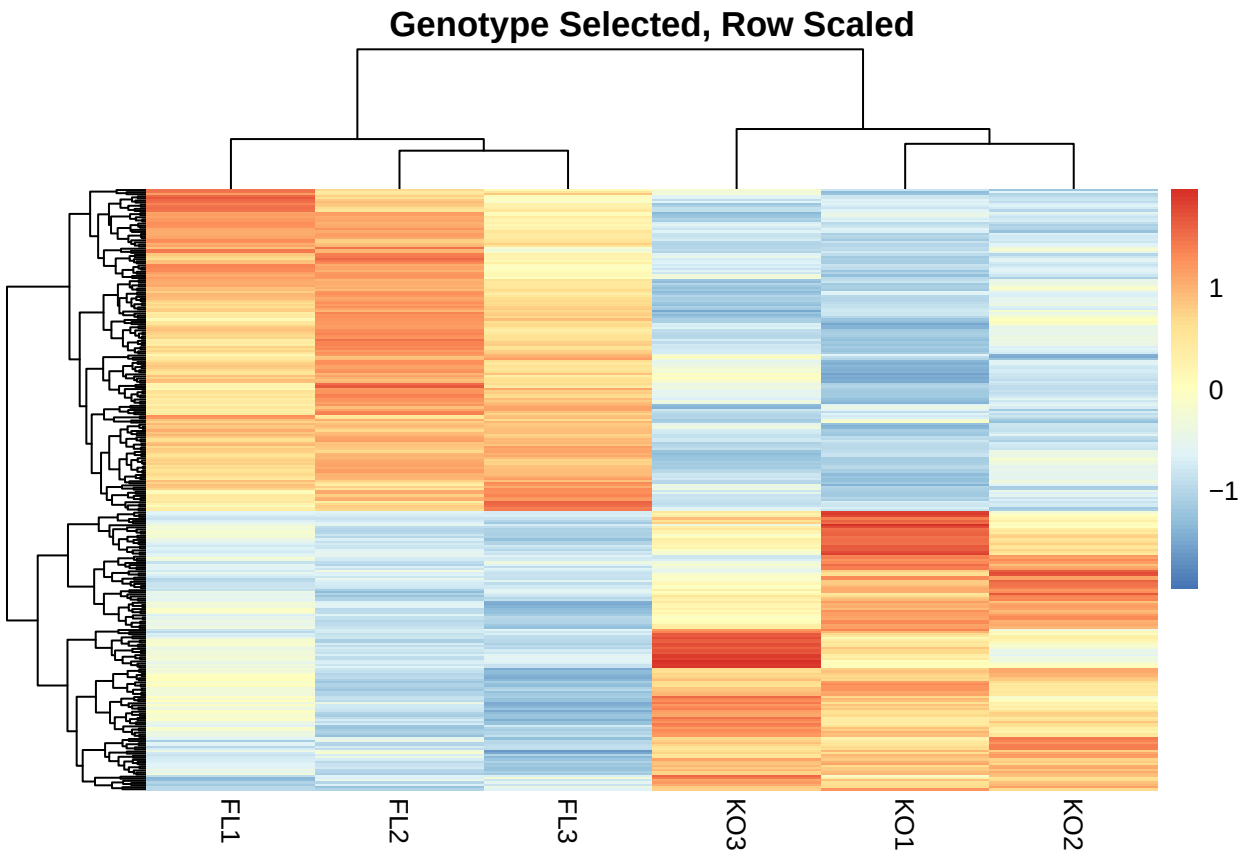
```
## ENSMUSG00000094694   32.3994       -5.45457   1.08528  -5.02596 5.00933e-07
##                           padj
##                      <numeric>
## ENSMUSG00000096410 0.007611827
## ENSMUSG00000094546 0.000400859
## ENSMUSG00000094689 0.000500484
## ENSMUSG00000093861 0.000217795
## ENSMUSG00000094420 0.038171081
## ENSMUSG00000094694 0.000232384
```

Make heatmaps from each of the output tables- selected top 50 or all using fold change and FDR.

```
top50 <-rownames(result_genotype_selected)[1:50]
#make a heatmap for the top 50
pheatmap(assay(rld_rnaseq)[top50,],scale="row",show_rownames=F,main="RLD values, Row Scaled, Top 50")
```

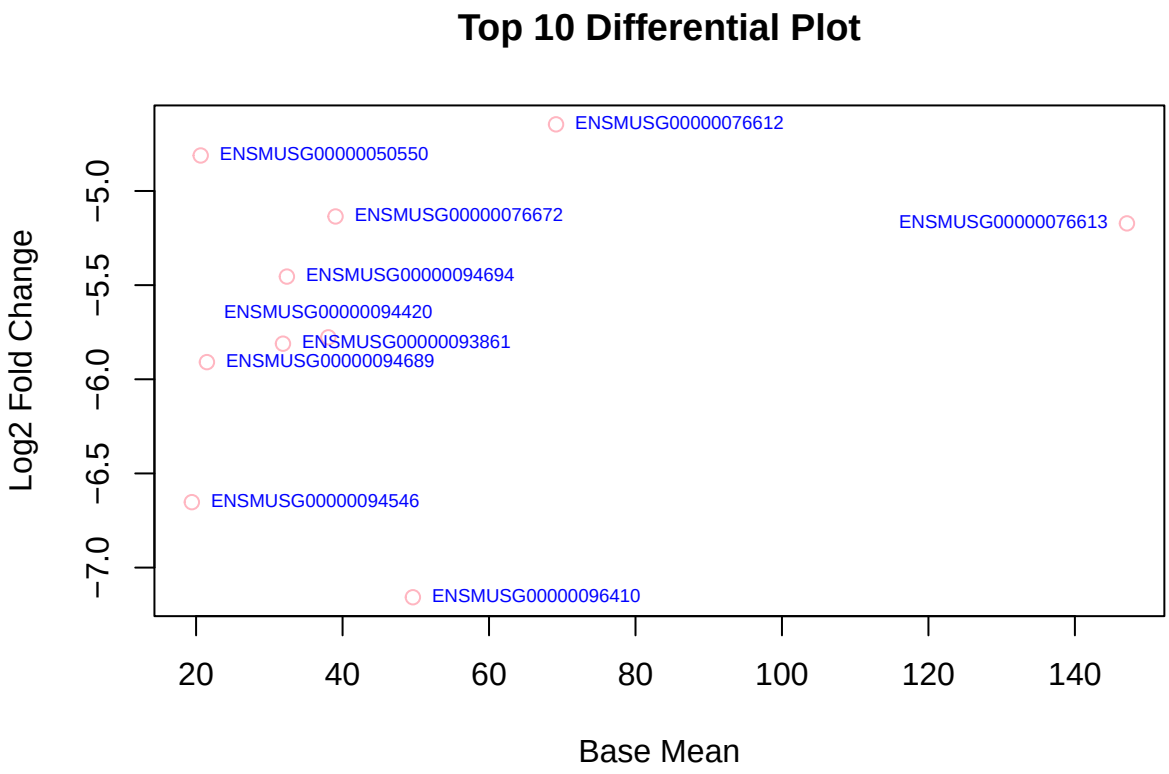



```
#all
pheatmap(assay(rld_rnaseq)[rownames(result_genotype_selected),],scale="row",
         show_rownames=F,main="Genotype Selected, Row Scaled")
```




```
#I did not use the rest of this script as my study only had one condition: the genotype.
#and the cell lines
#result_cell_selected <- subset(result_cell_line, padj < 0.05)
#result_cell_selected <-result_cell_selected[order(abs(result_cell_selected$log2FoldChange),decreasing =
#head(result_cell_selected)
#top50c <-rownames(result_cell_selected)[1:50]
#make a heatmap for the top 50
#pheatmap(assay(rld_rnaseq)[top50c,],scale="row",show_rownames=F, main="Cell Line Selected, Row Scaled,
#all
#pheatmap(assay(rld_rnaseq)[rownames(result_cell_selected),],scale="row",show_rownames=F,main="Cell Lin
```

```
#I wanted to see the top 10 list of fold change
top10 <- rownames(result_genotype_selected)[1:10]
log2dchange <- result_genotype_selected$log2FoldChange
basemean <- result_genotype_selected$baseMean
log2dchange <-c(log2dchange[1:10])
basemean <-c(basemean[1:10])
log2dchange<- as.data.frame(log2dchange)
log2dchange<-cbind(top10, log2dchange, basemean)
#Code written with the help of
#https://stackoverflow.com/questions/13229546/how-can-i-label-points-in-this-scatterplot
#and
#https://stackoverflow.com/questions/27368621/using-a-pos-vector-with-text-in-r
plot(log2dchange$basemean, log2dchange$log2dchange, main="Top 10 Differential Plot", xlab="Base Mean",
text(log2dchange$basemean, log2dchange$log2dchange, labels=log2dchange$top10, cex= 0.6, pos=list(4,4,4,
```



Build Gene Annotation. A serialised result "mart" has been generated in case Ensembl is unavailable

```
#Download Ensembl annotation using BiomaRt and rename the samples
library(biomaRt)
#UK ensembl is being updated so we use a USA mirror, "useast.ensembl.org"
ensembl_host <-"https://www.ensembl.org"
head(biomaRt::listMarts(host = ensembl_host), 15)
```

```
##                biomart                version
## 1 ENSEMBL_MART_ENSEMBL      Ensembl Genes 113
## 2   ENSEMBL_MART_MOUSE      Mouse strains 113
## 3     ENSEMBL_MART_SNP  Ensembl Variation 113
## 4 ENSEMBL_MART_FUNCGEN Ensembl Regulation 113
```

```
head(biomaRt::listAttributes(biomaRt::useDataset(dataset = "mmusculus_gene_ensembl",mart = useMart("ENS
```

```
##                              name                          description
## 1                 ensembl_gene_id                    Gene stable ID
## 2         ensembl_gene_id_version             Gene stable ID version
## 3           ensembl_transcript_id               Transcript stable ID
## 4    ensembl_transcript_id_version          Transcript stable ID version
## 5               ensembl_peptide_id                   Protein stable ID
## 6      ensembl_peptide_id_version              Protein stable ID version
## 7                 ensembl_exon_id                    Exon stable ID
## 8                     description                Gene description
## 9                 chromosome_name           Chromosome/scaffold name
## 10                 start_position                  Gene start (bp)
## 11                   end_position                    Gene end (bp)
```

```
## 12                          strand                                     Strand
## 13                            band                               Karyotype band
## 14                 transcript_start                        Transcript start (bp)
## 15                   transcript_end                          Transcript end (bp)
## 16          transcription_start_site               Transcription start site (TSS)
## 17                transcript_length Transcript length (including UTRs and CDS)
## 18                   transcript_tsl               Transcript support level (TSL)
## 19          transcript_gencode_basic                     GENCODE basic annotation
## 20                transcript_appris                             APPRIS annotation
## 21           transcript_is_canonical                             Ensembl Canonical
## 22                external_gene_name                                     Gene name
## 23              external_gene_source                          Source of gene name
## 24          external_transcript_name                              Transcript name
## 25 external_transcript_source_name                    Source of transcript name
## 26                  transcript_count                               Transcript count
## 27       percentage_gene_gc_content                             Gene % GC content
## 28                       gene_biotype                                     Gene type
## 29                 transcript_biotype                               Transcript type
## 30                             source                                 Source (gene)
## 31                 transcript_source                          Source (transcript)
## 32                            version                               Version (gene)
## 33                 transcript_version                        Version (transcript)
## 34                    peptide_version                           Version (protein)
## 35                  external_synonym                                 Gene Synonym
## 36             phenotype_description                       Phenotype description
## 37                         source_name                                  Source name
## 38                  study_external_id                   Study external reference
## 39                        strain_name                                  Strain name
## 40                       strain_gender                                Strain gender
##            page
## 1  feature_page
## 2  feature_page
## 3  feature_page
## 4  feature_page
## 5  feature_page
## 6  feature_page
## 7  feature_page
## 8  feature_page
## 9  feature_page
## 10 feature_page
## 11 feature_page
## 12 feature_page
## 13 feature_page
## 14 feature_page
## 15 feature_page
## 16 feature_page
## 17 feature_page
## 18 feature_page
## 19 feature_page
## 20 feature_page
## 21 feature_page
## 22 feature_page
## 23 feature_page
## 24 feature_page
```

```
## 25 feature_page
## 26 feature_page
## 27 feature_page
## 28 feature_page
## 29 feature_page
## 30 feature_page
## 31 feature_page
## 32 feature_page
## 33 feature_page
## 34 feature_page
## 35 feature_page
## 36 feature_page
## 37 feature_page
## 38 feature_page
## 39 feature_page
## 40 feature_page
```

```
mart <- biomaRt::useDataset(dataset = "mmusculus_gene_ensembl",mart    = useMart("ENSEMBL_MART_ENSEMBL"
resultAnnot <- biomaRt::getBM(values=rownames(dds_rnaseq),attributes = c("ensembl_gene_id","external_ge
#I wanted to work on the copy of annots file, just incase if something went wrong, I did not want to ma
annot_copy<- resultAnnot

getwd()
```

```
## [1] "/home/s2685584/ICA_functional"
```

Merge Annotation with Input Data

```
#merge with input data
names <- resultAnnot[,1]
resultAnnot <- as.data.frame(resultAnnot)
rownames(resultAnnot) = names
idx<-match(rownames(dds_rnaseq),rownames(resultAnnot))
#make sure annotation is in same order
all(rownames(dds_rnaseq) == rownames(resultAnnot))
```

```
## [1] FALSE
```

```
grr<-resultAnnot[match(rownames(dds_rnaseq), resultAnnot$ensembl_gene_id),]
all(rownames(dds_rnaseq) == rownames(grr))
```

```
## [1] TRUE
```

```
resultAnnot <-grr
all(rownames(dds_rnaseq) == rownames(resultAnnot))
```

```
## [1] TRUE
```

```
#make the nice names
nice_names<-paste(resultAnnot$ensembl_gene_id,resultAnnot$external_gene_name, sep = '_')
resultAnnot$nice_names <-nice_names
head(resultAnnot)
```

```
##                       ensembl_gene_id external_gene_name chromosome_name
## ENSMUSG00000051285 ENSMUSG00000051285             Pcmtd1               1
## ENSMUSG00000103509 ENSMUSG00000103509            Gm38372               1
## ENSMUSG00000057363 ENSMUSG00000057363               Uxs1               1
## ENSMUSG00000047216 ENSMUSG00000047216              Cdh19               1
```

```
## ENSMUSG00000038702 ENSMUSG00000038702               Dsel               1
## ENSMUSG00000061024 ENSMUSG00000061024               Rrs1               1
##                    start_position end_position
## ENSMUSG00000051285        7159144      7243852
## ENSMUSG00000103509        7218334      7222361
## ENSMUSG00000057363       43786126     43866960
## ENSMUSG00000047216      110816056    110905314
## ENSMUSG00000038702      111786432    111792648
## ENSMUSG00000061024        9615633      9617680
##
## ENSMUSG00000051285 protein-L-isoaspartate (D-aspartate) O-methyltransferase domain containing 1 [Sou
## ENSMUSG00000103509                                                          predicted gene, 38372 [Sou
## ENSMUSG00000057363                                                UDP-glucuronate decarboxylase 1 [Sou
## ENSMUSG00000047216                                                            cadherin 19, type 2 [Sou
## ENSMUSG00000038702                                                dermatan sulfate epimerase-like [Sou
## ENSMUSG00000061024                                                ribosome biogenesis regulator 1 [Sou
##                    strand                 nice_names
## ENSMUSG00000051285      1  ENSMUSG00000051285_Pcmtd1
## ENSMUSG00000103509      1 ENSMUSG00000103509_Gm38372
## ENSMUSG00000057363     -1    ENSMUSG00000057363_Uxs1
## ENSMUSG00000047216     -1   ENSMUSG00000047216_Cdh19
## ENSMUSG00000038702     -1    ENSMUSG00000038702_Dsel
## ENSMUSG00000061024      1    ENSMUSG00000061024_Rrs1
```

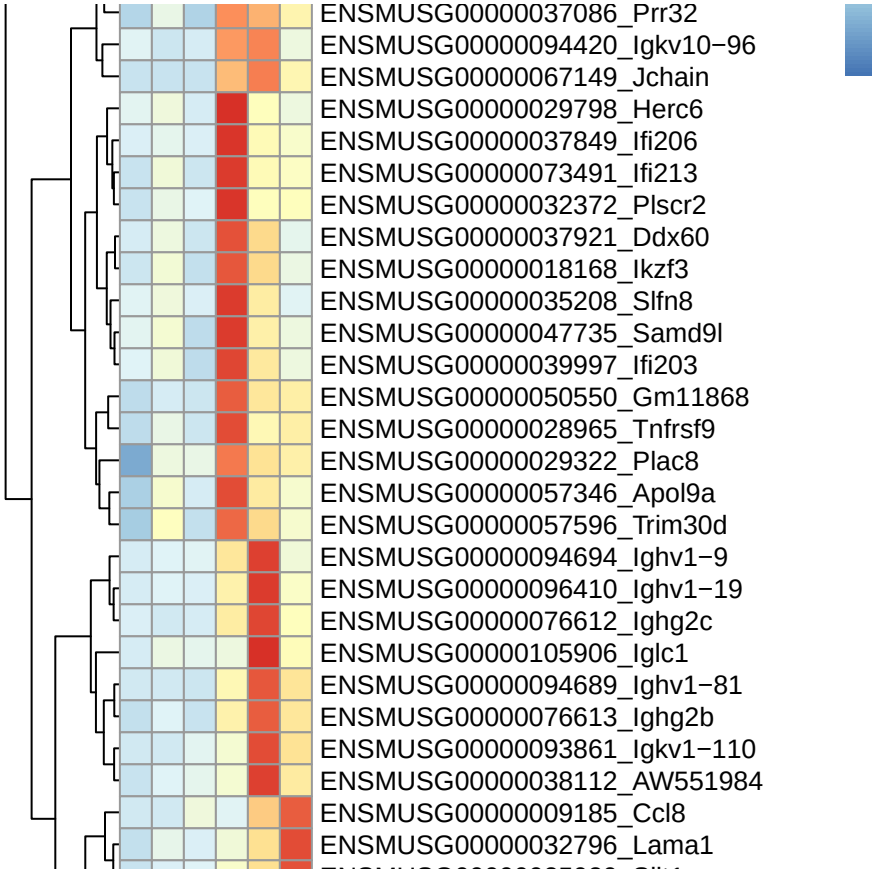
```
all(rownames(dds_rnaseq) == rownames(resultAnnot))
```

```
## [1] TRUE
```

```
#check names
rld_rnaseq <- rlog(dds_rnaseq, blind = TRUE)#
idx2 <-match(rownames(result_genotype_selected)[1:50],rownames(dds_rnaseq))
plotme <-(rld_rnaseq)[rownames(result_genotype_selected)[1:50],]
rownames(plotme)<-resultAnnot$nice_names[idx2]
write.csv(result_genotype, "results_genotype.csv", sep="\t")
```

```
## Warning in write.csv(result_genotype, "results_genotype.csv", sep = "\t"):
## attempt to set 'sep' ignored
```

```
#make heatmap with candidate genes
png(filename="heatmap_candidates_ICA.png",width=800, height=1000)
pheatmap(assay(plotme),scale="row",main="Candidate Genes",
         fontsize_row = 10,cellheight =12, cellwidth =12,treeheight_row = 40, treeheight_col = 40)
```




```
dev.off()
```

```
## png
##   3
```

```
#pheatmap(assay(plotme),scale="row",fontsize_row = 10,cellheight =12, cellwidth =12,treeheight_row = 40
```

```
``` r
## Final Results Table
#Since we have annotation that matches gene names to candidates we can also annotate the results table
#from differential expression
#Code beyond this block were taken from Example Analysis Extra ICA document
#of Functional Genomic Technologies class material.
```

```
#I edited the code according to my workspace and my dataset
#this has the results ordered
has_annot <- !annot_copy$gene==""

#before
head(resultAnnot,50)
```

```
##                      ensembl_gene_id external_gene_name chromosome_name
## ENSMUSG00000051285 ENSMUSG00000051285             Pcmtd1               1
## ENSMUSG00000103509 ENSMUSG00000103509            Gm38372               1
## ENSMUSG00000057363 ENSMUSG00000057363               Uxs1               1
## ENSMUSG00000047216 ENSMUSG00000047216              Cdh19               1
## ENSMUSG00000038702 ENSMUSG00000038702               Dsel               1
```
```

```
## ENSMUSG00000061024 ENSMUSG00000061024              Rrs1               1
## ENSMUSG00000025911 ENSMUSG00000025911            Adhfe1               1
## ENSMUSG00000057335 ENSMUSG00000057335            Cep170               1
## ENSMUSG00000045210 ENSMUSG00000045210            Vcpip1               1
## ENSMUSG00000041763 ENSMUSG00000041763              Tpp2               1
## ENSMUSG00000026504 ENSMUSG00000026504           Sdccag8               1
## ENSMUSG00000047343 ENSMUSG00000047343          Mettl21c               1
## ENSMUSG00000103251 ENSMUSG00000103251           Gm37519               1
## ENSMUSG00000026483 ENSMUSG00000026483            Niban1               1
## ENSMUSG00000019699 ENSMUSG00000019699              Akt3               1
## ENSMUSG00000063659 ENSMUSG00000063659            Zbtb18               1
## ENSMUSG00000026049 ENSMUSG00000026049              Tex30              1
## ENSMUSG00000043019 ENSMUSG00000043019              Edem3              1
## ENSMUSG00000032666 ENSMUSG00000032666     1700025G04Rik               1
## ENSMUSG00000015961 ENSMUSG00000015961              Adss2              1
## ENSMUSG00000014980 ENSMUSG00000014980            Tsen15               1
## ENSMUSG00000026235 ENSMUSG00000026235              Epha4              1
## ENSMUSG00000032649 ENSMUSG00000032649          Colgalt2               1
## ENSMUSG00000026047 ENSMUSG00000026047           Poglut2               1
## ENSMUSG00000041684 ENSMUSG00000041684              Bivm               1
## ENSMUSG00000026502 ENSMUSG00000026502             Desi2               1
## ENSMUSG00000026048 ENSMUSG00000026048             Ercc5               1
## ENSMUSG00000026245 ENSMUSG00000026245             Farsb               1
## ENSMUSG00000090486 ENSMUSG00000090486          BC035947               1
## ENSMUSG00000026482 ENSMUSG00000026482              Rgl1               1
## ENSMUSG00000012187 ENSMUSG00000012187            Mogat1               1
## ENSMUSG00000008475 ENSMUSG00000008475             Arpc5               1
## ENSMUSG00000056870 ENSMUSG00000056870             Gulp1               1
## ENSMUSG00000039630 ENSMUSG00000039630            Hnrnpu               1
## ENSMUSG00000104279 ENSMUSG00000104279           Gm38300               1
## ENSMUSG00000032883 ENSMUSG00000032883             Acsl3               1
## ENSMUSG00000026480 ENSMUSG00000026480              Ncf2               1
## ENSMUSG00000026494 ENSMUSG00000026494            Kif26b               1
## ENSMUSG00000025915 ENSMUSG00000025915              Sgk3               1
## ENSMUSG00000079470 ENSMUSG00000079470            Utp14b               1
## ENSMUSG00000047330 ENSMUSG00000047330             Kcne4               1
## ENSMUSG00000055067 ENSMUSG00000055067             Smyd3               1
## ENSMUSG00000054702 ENSMUSG00000054702             Ap1s3               1
## ENSMUSG00000026492 ENSMUSG00000026492             Tfb2m               1
## ENSMUSG00000038949 ENSMUSG00000038949              Cnst               1
## ENSMUSG00000026043 ENSMUSG00000026043            Col3a1               1
## ENSMUSG00000038936 ENSMUSG00000038936            Sccpdh               1
## ENSMUSG00000042772 ENSMUSG00000042772              Smg7               1
## ENSMUSG00000026042 ENSMUSG00000026042            Col5a2               1
## ENSMUSG00000104476 ENSMUSG00000104476           Gm38211               1
##                    start_position end_position
## ENSMUSG00000051285        7159144      7243852
## ENSMUSG00000103509        7218334      7222361
## ENSMUSG00000057363       43786126     43866960
## ENSMUSG00000047216      110816056    110905314
## ENSMUSG00000038702      111786432    111792648
## ENSMUSG00000061024        9615633      9617680
## ENSMUSG00000025911        9618173      9650898
## ENSMUSG00000057335      176561219    176641633
```

```
## ENSMUSG00000045210        9788847      9818607
## ENSMUSG00000041763       43972807     44042160
## ENSMUSG00000026504      176642226    176848003
## ENSMUSG00000047343       44048568     44059194
## ENSMUSG00000103251      176801802    176804011
## ENSMUSG00000026483      151446937    151597690
## ENSMUSG00000019699      176847639    177085769
## ENSMUSG00000063659      177269917    177278330
## ENSMUSG00000026049       44125773     44141601
## ENSMUSG00000043019      151631122    151697802
## ENSMUSG00000032666      151728154    151965876
## ENSMUSG00000015961      177590528    177624077
## ENSMUSG00000014980      152246486    152262439
## ENSMUSG00000026235       77343822     77491725
## ENSMUSG00000032649      152275581    152386446
## ENSMUSG00000026047       44145706     44157968
## ENSMUSG00000041684       44158117     44183930
## ENSMUSG00000026502      178014983    178084867
## ENSMUSG00000026048       44186904     44220420
## ENSMUSG00000026245       78394612     78465534
## ENSMUSG00000090486       78473663     78488795
## ENSMUSG00000026482      152392511    152642102
## ENSMUSG00000012187       78487628     78514810
## ENSMUSG00000008475      152642293    152651348
## ENSMUSG00000056870       44590671     44835998
## ENSMUSG00000039630      178148673    178165362
## ENSMUSG00000104279      178167166    178169035
## ENSMUSG00000032883       78635542     78685460
## ENSMUSG00000026480      152675945    152712742
## ENSMUSG00000026494      178356690    178766765
## ENSMUSG00000025915        9868332      9971070
## ENSMUSG00000079470       78635755     78649229
## ENSMUSG00000047330       78794475     78797745
## ENSMUSG00000055067      178779525    179345606
## ENSMUSG00000054702       79584593     79649689
## ENSMUSG00000026492      179355620    179373832
## ENSMUSG00000038949      179373935    179455043
## ENSMUSG00000026043       45350698     45388866
## ENSMUSG00000038936      179495775    179514754
## ENSMUSG00000042772      152712746    152778397
## ENSMUSG00000026042       45413481     45542442
## ENSMUSG00000104476       45524996     45532212
##
## ENSMUSG00000051285         protein-L-isoaspartate (D-aspartate) O-methyltransferase domain containing
## ENSMUSG00000103509                                                                predicted gene, 383
## ENSMUSG00000057363                                                  UDP-glucuronate decarboxylase
## ENSMUSG00000047216                                                              cadherin 19, type
## ENSMUSG00000038702                                                  dermatan sulfate epimerase-li
## ENSMUSG00000061024                                                  ribosome biogenesis regulator
## ENSMUSG00000025911                                        alcohol dehydrogenase, iron containing
## ENSMUSG00000057335                                                        centrosomal protein 1
## ENSMUSG00000045210                valosin containing protein (p97)/p47 complex interacting protein
## ENSMUSG00000041763                                                        tripeptidyl peptidase
## ENSMUSG00000026504                                  serologically defined colon cancer antigen
```

```
## ENSMUSG00000047343                                                    methyltransferase 21C, AARS1 lysi
## ENSMUSG00000103251                                                                 predicted gene, 375
## ENSMUSG00000026483                                                       niban apoptosis regulatoi
## ENSMUSG00000019699                                                   thymoma viral proto-oncogene
## ENSMUSG00000063659                                          zinc finger and BTB domain containing
## ENSMUSG00000026049                                                              testis expressed
## ENSMUSG00000043019                                  ER degradation enhancer, mannosidase alpha-like
## ENSMUSG00000032666                                                    RIKEN cDNA 1700025G04 ge
## ENSMUSG00000015961                                                     adenylosuccinate syntha
## ENSMUSG00000014980                                       tRNA splicing endonuclease subunit
## ENSMUSG00000026235                                                                  Eph recepto
## ENSMUSG00000032649                                  collagen beta(1-O)galactosyltransferase
## ENSMUSG00000026047                                           protein O-glucosyltransferase
## ENSMUSG00000041684                        basic, immunoglobulin-like variable motif containi
## ENSMUSG00000026502                                              desumoylating isopeptidase
## ENSMUSG00000026048 excision repair cross-complementing rodent repair deficiency, complementation grou
## ENSMUSG00000026245                                  phenylalanyl-tRNA synthetase, beta subun
## ENSMUSG00000090486                                                        cDNA sequence BC0359
## ENSMUSG00000026482                       ral guanine nucleotide dissociation stimulator,-like
## ENSMUSG00000012187                                    monoacylglycerol O-acyltransferase
## ENSMUSG00000008475                                 actin related protein 2/3 complex, subunit
## ENSMUSG00000056870                                GULP, engulfment adaptor PTB domain containing
## ENSMUSG00000039630                                heterogeneous nuclear ribonucleoprotein
## ENSMUSG00000104279                                                      predicted gene, 383
## ENSMUSG00000032883                          acyl-CoA synthetase long-chain family member
## ENSMUSG00000026480                                                neutrophil cytosolic fact
## ENSMUSG00000026494                                                  kinesin family member 1
## ENSMUSG00000025915                                  serum/glucocorticoid regulated kinase
## ENSMUSG00000079470                                UTP14B small subunit processome compone
## ENSMUSG00000047330                  potassium voltage-gated channel, Isk-related subfamily, gene
## ENSMUSG00000055067                                       SET and MYND domain containing
## ENSMUSG00000054702                          adaptor-related protein complex AP-1, sigma
## ENSMUSG00000026492                                  transcription factor B2, mitochondr
## ENSMUSG00000038949                                    consortin, connexin sorting prote
## ENSMUSG00000026043                                                collagen, type III, alp
## ENSMUSG00000038936                                 saccharopine dehydrogenase (putativ
## ENSMUSG00000042772                               SMG7 nonsense mediated mRNA decay fact
## ENSMUSG00000026042                                                  collagen, type V, alp
## ENSMUSG00000104476                                                      predicted gene, 383
##                     strand                      nice_names
## ENSMUSG00000051285      1        ENSMUSG00000051285_Pcmtd1
## ENSMUSG00000103509      1       ENSMUSG00000103509_Gm38372
## ENSMUSG00000057363     -1          ENSMUSG00000057363_Uxs1
## ENSMUSG00000047216     -1         ENSMUSG00000047216_Cdh19
## ENSMUSG00000038702     -1          ENSMUSG00000038702_Dsel
## ENSMUSG00000061024      1          ENSMUSG00000061024_Rrs1
## ENSMUSG00000025911      1        ENSMUSG00000025911_Adhfe1
## ENSMUSG00000057335     -1        ENSMUSG00000057335_Cep170
## ENSMUSG00000045210     -1        ENSMUSG00000045210_Vcpip1
## ENSMUSG00000041763      1          ENSMUSG00000041763_Tpp2
## ENSMUSG00000026504      1       ENSMUSG00000026504_Sdccag8
## ENSMUSG00000047343     -1      ENSMUSG00000047343_Mettl21c
## ENSMUSG00000103251      1       ENSMUSG00000103251_Gm37519
## ENSMUSG00000026483      1        ENSMUSG00000026483_Niban1
```

```
## ENSMUSG00000019699     -1          ENSMUSG00000019699_Akt3
## ENSMUSG00000063659      1        ENSMUSG00000063659_Zbtb18
## ENSMUSG00000026049     -1         ENSMUSG00000026049_Tex30
## ENSMUSG00000043019      1         ENSMUSG00000043019_Edem3
## ENSMUSG00000032666     -1 ENSMUSG00000032666_1700025G04Rik
## ENSMUSG00000015961     -1         ENSMUSG00000015961_Adss2
## ENSMUSG00000014980     -1        ENSMUSG00000014980_Tsen15
## ENSMUSG00000026235     -1         ENSMUSG00000026235_Epha4
## ENSMUSG00000032649      1      ENSMUSG00000032649_Colgalt2
## ENSMUSG00000026047     -1       ENSMUSG00000026047_Poglut2
## ENSMUSG00000041684      1          ENSMUSG00000041684_Bivm
## ENSMUSG00000026502      1         ENSMUSG00000026502_Desi2
## ENSMUSG00000026048      1         ENSMUSG00000026048_Ercc5
## ENSMUSG00000026245     -1         ENSMUSG00000026245_Farsb
## ENSMUSG00000090486     -1      ENSMUSG00000090486_BC035947
## ENSMUSG00000026482     -1          ENSMUSG00000026482_Rgl1
## ENSMUSG00000012187      1        ENSMUSG00000012187_Mogat1
## ENSMUSG00000008475      1         ENSMUSG00000008475_Arpc5
## ENSMUSG00000056870      1         ENSMUSG00000056870_Gulp1
## ENSMUSG00000039630     -1        ENSMUSG00000039630_Hnrnpu
## ENSMUSG00000104279      1       ENSMUSG00000104279_Gm38300
## ENSMUSG00000032883      1         ENSMUSG00000032883_Acsl3
## ENSMUSG00000026480      1          ENSMUSG00000026480_Ncf2
## ENSMUSG00000026494      1        ENSMUSG00000026494_Kif26b
## ENSMUSG00000025915      1          ENSMUSG00000025915_Sgk3
## ENSMUSG00000079470      1        ENSMUSG00000079470_Utp14b
## ENSMUSG00000047330      1         ENSMUSG00000047330_Kcne4
## ENSMUSG00000055067     -1         ENSMUSG00000055067_Smyd3
## ENSMUSG00000054702     -1         ENSMUSG00000054702_Ap1s3
## ENSMUSG00000026492     -1         ENSMUSG00000026492_Tfb2m
## ENSMUSG00000038949      1          ENSMUSG00000038949_Cnst
## ENSMUSG00000026043      1        ENSMUSG00000026043_Col3a1
## ENSMUSG00000038936      1        ENSMUSG00000038936_Sccpdh
## ENSMUSG00000042772     -1          ENSMUSG00000042772_Smg7
## ENSMUSG00000026042     -1        ENSMUSG00000026042_Col5a2
## ENSMUSG00000104476     -1       ENSMUSG00000104476_Gm38211
```

```
result_genotype <-results(dds_rnaseq,contrast=c("Genotype","floxed","WBSCR16_KO"))
idx2 <-match(rownames(result_genotype),annot_copy$ensembl_gene_id)
rownames(result_genotype)<-annot_copy$external_gene_name[idx2]
#after
print(result_genotype[1:50,])
```

```
## log2 fold change (MLE): Genotype floxed vs WBSCR16_KO
## Wald test p-value: Genotype floxed vs WBSCR16_KO
## DataFrame with 50 rows and 6 columns
##           baseMean log2FoldChange     lfcSE        stat     pvalue      padj
##          <numeric>      <numeric> <numeric>   <numeric>  <numeric> <numeric>
## Pcmtd1    314.8884    0.006090424  0.400369  0.01521204  0.9878630  0.997285
## Gm38372    23.6041   -0.289636637  0.646882 -0.44774231  0.6543392  0.919220
## Uxs1      169.3759    0.000775135  0.258076  0.00300351  0.9976035  0.999382
## Cdh19      22.7042   -0.648586107  0.517694 -1.25283742  0.2102649  0.668292
## Dsel       47.3322   -0.878964331  0.507215 -1.73292106  0.0831097  0.454953
## ...             ...            ...       ...         ...        ...       ...
## Col3a1  46420.4670     -1.179687  0.421854   -2.796436 0.00516697  0.122131
```

```
## Sccpdh    185.2853       0.545819  0.374038    1.459261 0.14449323  0.578453
## Smg7     1292.2507      -0.116568  0.154721   -0.753412 0.45120248  0.854473
## Col5a2   3169.1751      -1.122338  0.408045   -2.750528 0.00594993  0.131236
## Gm38211    10.3485      -0.202829  0.942572   -0.215187 0.82962151        NA
```

```
#load into table viewer
#View(as.data.frame(result_treatment[1:50,]))
```

The source paper showed candidates in a volcano plot- this is to compare

Current row names are not useful here- we need to use genes but then will discard any without known names. This is OK here because we are exploring the annotation -genes without a gene name and so are no use for annotating the data.

We can still keep unannoted genes as candidates!

```
#Now we have gene names so we can plot the remaining genes in a volcano plot

#there were highlighted as interesting in the paper
paper_genes <-c("mt-Rnr2","mt-Rnr1","Eral1","Sdhb", "Cox4i1","Cox4i2", "Mrps16", "mt-Nd3","Rcc1l")
#just check they're measured in this annotation
if(sum(paper_genes %in% rownames(result_genotype))==length(paper_genes)){
  print("all genes found in data")
}#you'd just include a subset to label if some genes were missing- here they are all present
```

```
## [1] "all genes found in data"
```

```
library(EnhancedVolcano)
```

```
## Loading required package: ggrepel
```

```
#The paper found...
#for volvano see
#https://bioconductor.org/packages/devel/bioc/vignettes/EnhancedVolcano/inst/doc/EnhancedVolcano.html
volcano_test <-EnhancedVolcano(result_genotype,
    title="Volcano Plot of Gene Expression, with highlighted genes",
    lab = rownames(result_genotype),
    x = 'log2FoldChange',
    y = 'padj',
    selectLab = paper_genes,
    xlab = bquote(~Log[2]~ 'fold change'),
    pCutoff = 10e-5,
    FCcutoff = 2.0,
    pointSize = 4.0,
    labSize = 6.0,
    labCol = 'black',
    labFace = 'bold',
    boxedLabels = TRUE,
    colAlpha = 4/5,
    legendPosition = 'right',
    legendLabSize = 10,
    legendIconSize = 4.0,
    drawConnectors = TRUE,
    widthConnectors = 1.0,
    colConnectors = 'black')

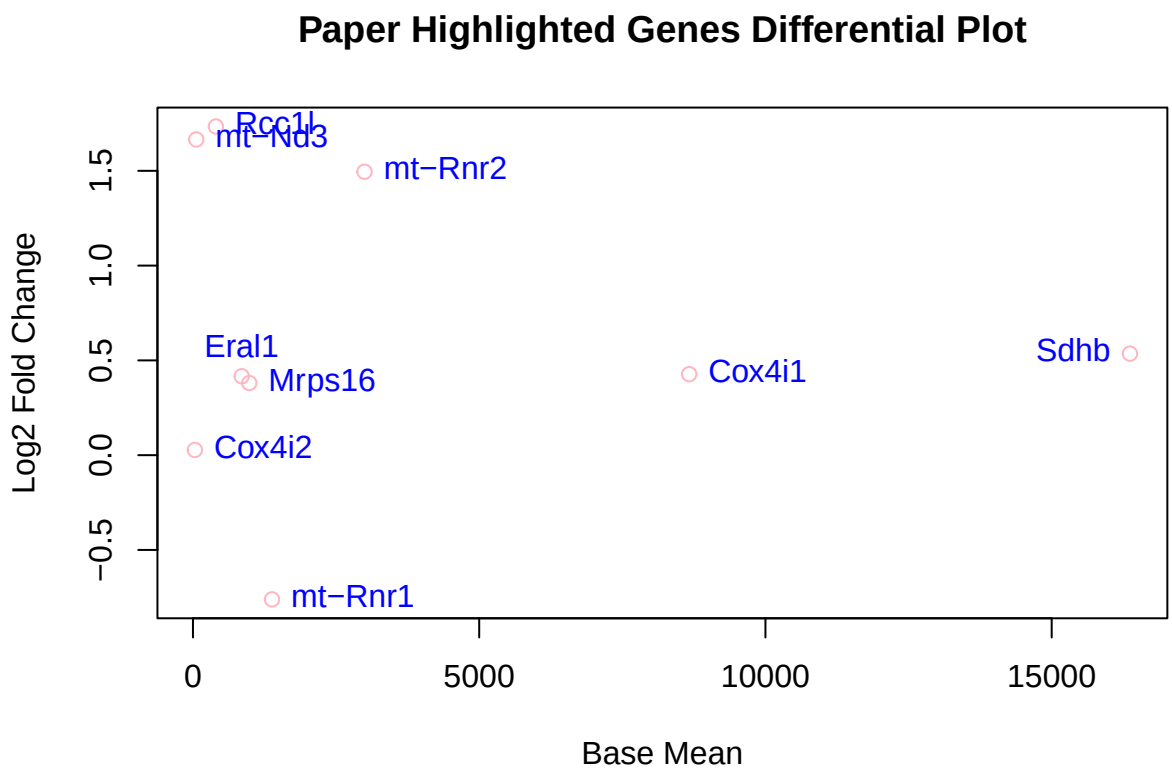
#plot(volcano_test,xlim=800,ylim=800) -> I did not run this as the window size was not enough
```

```
#(as stated in
#https://stackoverflow.com/questions/73935357/error-in-grid-callc-convert-x-as-integerwhatfrom-as-integ
#I just sent the plot to the file so I could have a plot as
#the code would stop working after the error.
png("volcano_test_3.png",width = 800, height = 800)
plot(volcano_test)
dev.off()
```

```
## pdf
##   2
```

```
#and look at the table
#View(as.data.frame(result_treatment[paper_genes,]))
```

```
#I wanted to see how much fold change the genes on the paper had
#Code written with the help of
#https://stackoverflow.com/questions/13229546/how-can-i-label-points-in-this-scatterplot
#and
#https://stackoverflow.com/questions/27368621/using-a-pos-vector-with-text-in-r
paper_genes_table <- as.data.frame(result_genotype[paper_genes,])
plot(paper_genes_table$baseMean, paper_genes_table$log2FoldChange, main="Paper Highlighted Genes Differ
xlab="Base Mean", ylab="Log2 Fold Change", col="lightpink")
text(paper_genes_table$baseMean, paper_genes_table$log2FoldChange, labels=rownames(paper_genes_table),
```



Additional Reading
http://bioconductor.org/packages/devel/bioc/vignettes/DESeq2/inst/doc/DESeq2.html

Session Information

``` r
sessionInfo()
```

```
## R version 4.4.3 (2025-02-28)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0
##
## locale:
##  [1] LC_CTYPE=en_GB.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_GB.UTF-8        LC_COLLATE=en_GB.UTF-8
##  [5] LC_MONETARY=en_GB.UTF-8    LC_MESSAGES=en_GB.UTF-8
##  [7] LC_PAPER=en_GB.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_GB.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/London
## tzcode source: system (glibc)
##
## attached base packages:
## [1] parallel  stats4    stats     graphics  grDevices utils     datasets
## [8] methods   base
##
## other attached packages:
##  [1] EnhancedVolcano_1.22.0      ggrepel_0.9.6
##  [3] biomaRt_2.60.1              RColorBrewer_1.1-3
##  [5] ggplot2_3.5.1               scatterplot3d_0.3-44
##  [7] pheatmap_1.0.12             QuasR_1.44.0
##  [9] Rbowtie_1.44.0              affy_1.82.0
## [11] DESeq2_1.44.0               SummarizedExperiment_1.34.0
## [13] Biobase_2.64.0              MatrixGenerics_1.16.0
## [15] matrixStats_1.4.1           GenomicRanges_1.56.2
## [17] GenomeInfoDb_1.40.1         IRanges_2.38.1
## [19] S4Vectors_0.42.1            BiocGenerics_0.50.0
## [21] gplots_3.2.0                edgeR_4.2.2
## [23] limma_3.60.6                Rsubread_2.18.0
##
## loaded via a namespace (and not attached):
##   [1] rstudioapi_0.17.1        jsonlite_1.8.9           magrittr_2.0.3
##   [4] GenomicFeatures_1.56.0   farver_2.1.2             rmarkdown_2.29
##   [7] BiocIO_1.14.0            zlibbioc_1.50.0          vctrs_0.6.5
##  [10] memoise_2.0.1            Rsamtools_2.20.0         RCurl_1.98-1.16
##  [13] htmltools_0.5.8.1        S4Arrays_1.4.1           progress_1.2.3
##  [16] curl_6.0.1               SparseArray_1.4.8        KernSmooth_2.23-24
##  [19] httr2_1.0.7              cachem_1.1.0             GenomicAlignments_1.40.0
##  [22] lifecycle_1.0.4          pkgconfig_2.0.3          Matrix_1.7-1
##  [25] R6_2.5.1                 fastmap_1.2.0            GenomeInfoDbData_1.2.12
##  [28] digest_0.6.37            colorspace_2.1-1         ShortRead_1.62.0
```

```
##  [31] AnnotationDbi_1.66.0     RSQLite_2.3.9            hwriter_1.3.2.1
##  [34] filelock_1.0.3           labeling_0.4.3           fansi_1.0.6
##  [37] httr_1.4.7               abind_1.4-8              compiler_4.4.3
##  [40] bit64_4.5.2              withr_3.0.2              BiocParallel_1.38.0
##  [43] DBI_1.2.3                rappdirs_0.3.3           DelayedArray_0.30.1
##  [46] rjson_0.2.23             gtools_3.9.5             caTools_1.18.3
##  [49] tools_4.4.3              glue_1.8.0               restfulr_0.0.15
##  [52] grid_4.4.3               generics_0.1.3           gtable_0.3.6
##  [55] BSgenome_1.72.0          preprocessCore_1.66.0    hms_1.1.3
##  [58] xml2_1.3.6               utf8_1.2.4               XVector_0.44.0
##  [61] pillar_1.9.0             stringr_1.5.1            dplyr_1.1.4
##  [64] BiocFileCache_2.12.0     lattice_0.22-6           rtracklayer_1.64.0
##  [67] bit_4.5.0.1              deldir_2.0-4             tidyselect_1.2.1
##  [70] locfit_1.5-9.10          Biostrings_2.72.1        knitr_1.49
##  [73] xfun_0.49                statmod_1.5.0            stringi_1.8.4
##  [76] UCSC.utils_1.0.0         yaml_2.3.10              evaluate_1.0.1
##  [79] codetools_0.2-20         interp_1.1-6             GenomicFiles_1.40.0
##  [82] tibble_3.2.1             BiocManager_1.30.25      cli_3.6.3
##  [85] affyio_1.74.0            munsell_0.5.1            Rcpp_1.0.13-1
##  [88] dbplyr_2.5.0             png_0.1-8                XML_3.99-0.17
##  [91] blob_1.2.4               prettyunits_1.2.0        latticeExtra_0.6-30
##  [94] jpeg_0.1-10              bitops_1.0-9             pwalign_1.0.0
##  [97] txdbmaker_1.0.1          VariantAnnotation_1.50.0 scales_1.3.0
## [100] purrr_1.0.2              crayon_1.5.3             rlang_1.1.4
## [103] KEGGREST_1.44.1
```